Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

INTRODUCTION

Jared Ortiz-Angulo Cantos

Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025

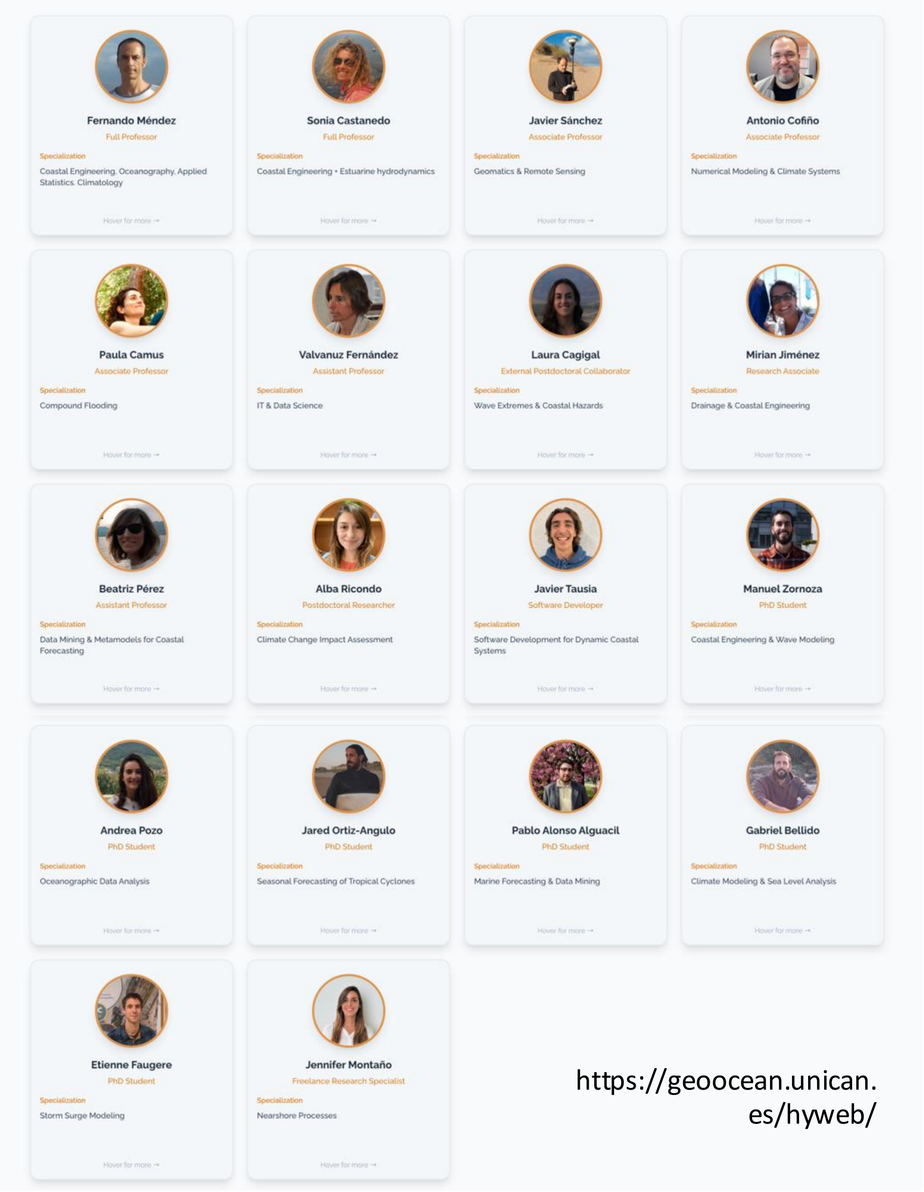
October 20th-23th, 2025

Introduction

UC UNIVERSIDAD DE CANTABRIA

GeOcean Geomatics and Ocean Engineering Group

- Coastal Flooding in a Changing Climate
- Hybrid Wave Downscaling Models
- Climate-based MetOcean Emulators
- Early Warning Systems
- Risk Management of Natural Disasters
- Hybrid Modeling of Tropical Cyclone-induced Waves



https://geoocean.unican.es/hyweb/

Introduction

About me

2015

PRACTICUM STUDENT
Fundación Instituto Hidráulica Ambiental del Cantabria
IH Cantabria. Universidad de Cantabria

2017- 2019

CONSULTANT
MCVALNERA
Santander, Cantabria

2019-2022

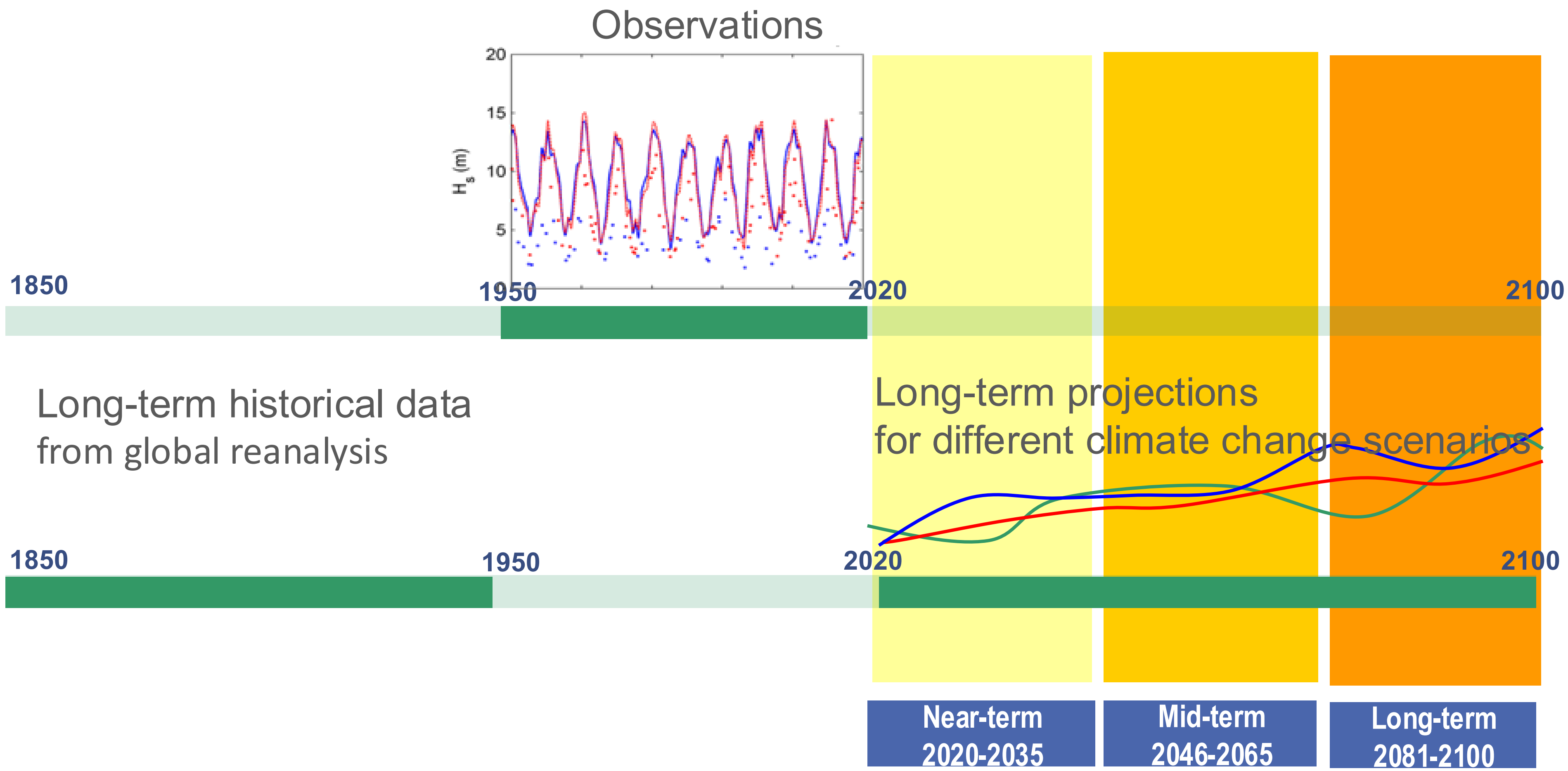
PROJECT MANAGER
MCVALNERA
Santander, Cantabria

2022 - NOW

RESEARCHER – PhD STUDENT
Grupo de Ingeniería Geomática y Oceanográfica
Universidad de Cantabria

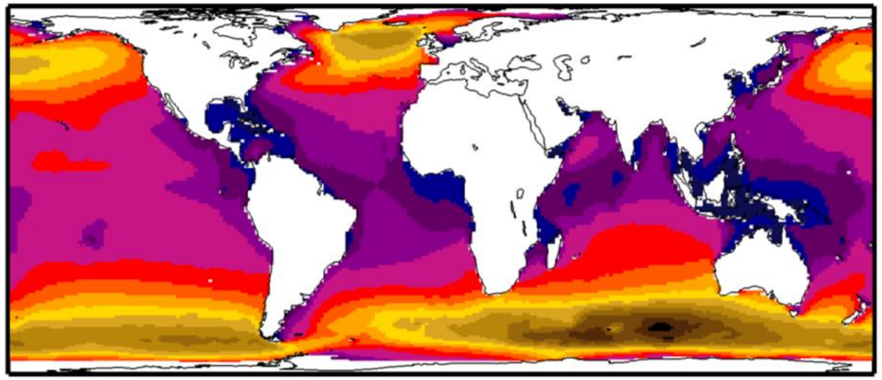
Introduction

Available data for Statistical downscaling approach based on weather typesx

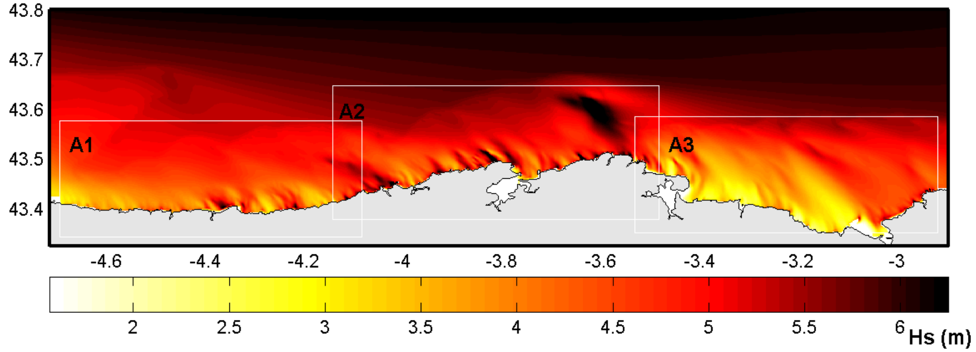


Introduction

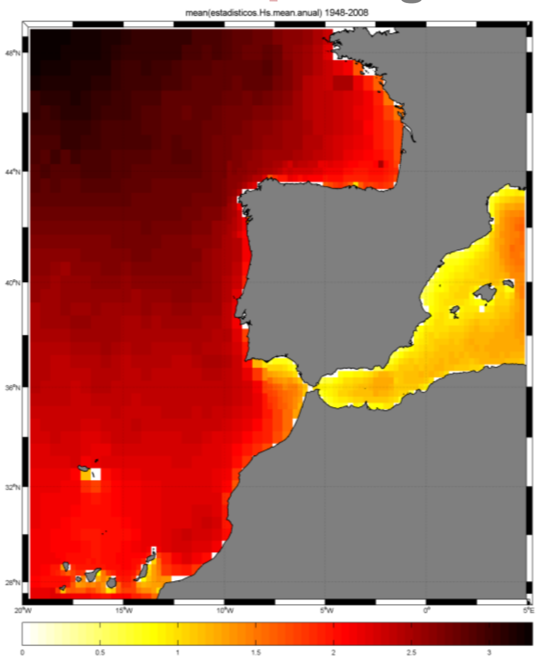
Global



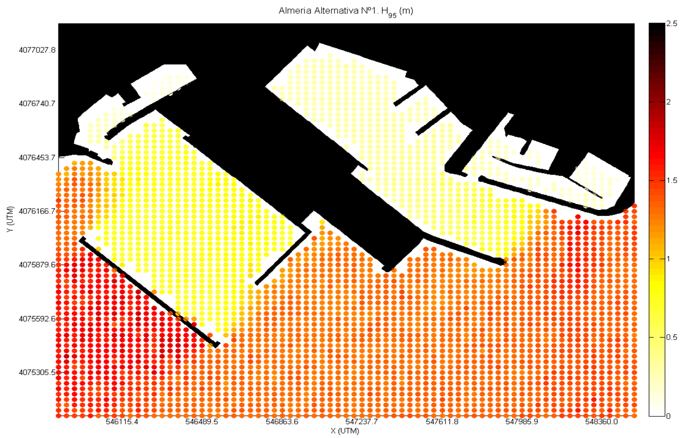
Local



Regional

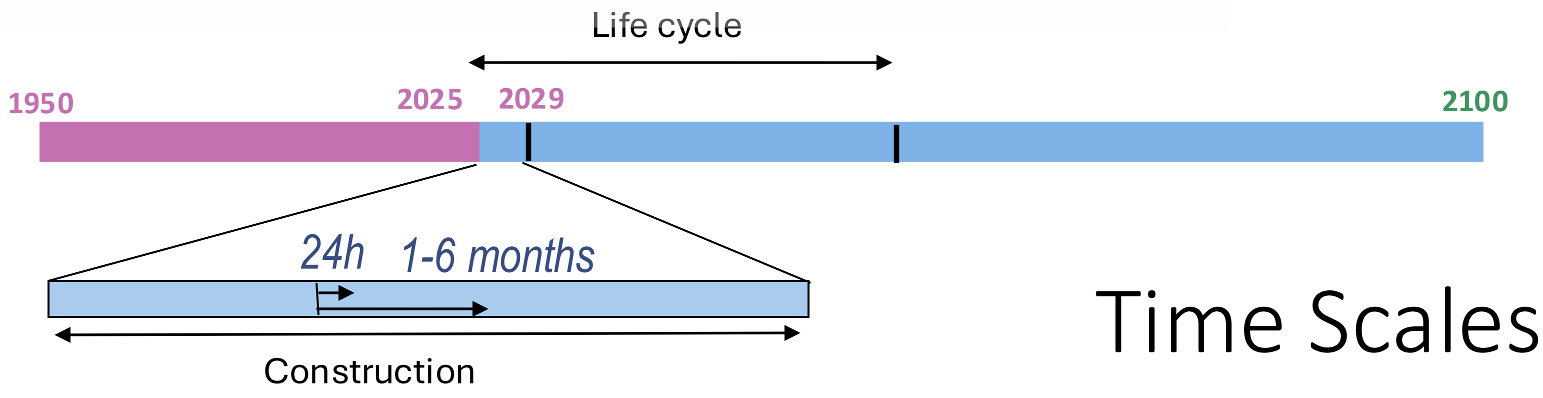


High Resolution



Spatial Scales

Introduction



Time Scales of Interest

Design phase:
- long-term
- very-long term

Construction phase:
- short-term
- medium-term

Operations and maintenance phase:
- short-term
- medium-term

Re-use/demolition/adaptation phase:
- short, medium, long and very-long term

Short-term
24-72 hours Forecasting marine dynamics

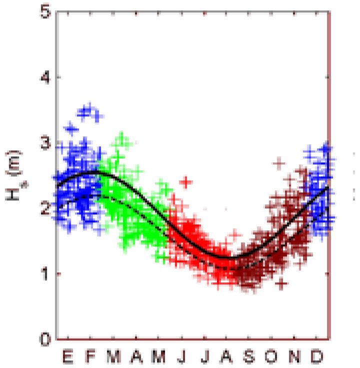
Medium term
Seasonal prediction (between 1 and 6 months)

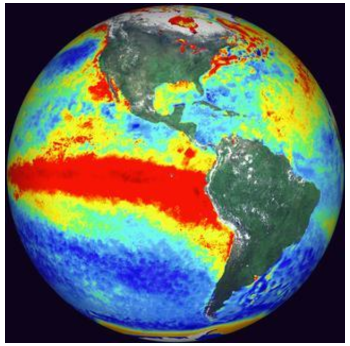
Long-term
Probabilistic analysis based on historical records

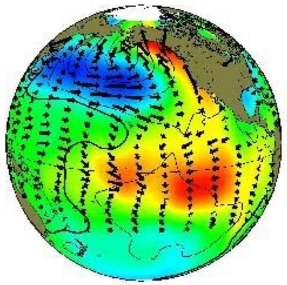
Very-long-term
Long-term trends and projection to IPCC scenarios

Introduction

Time Scales of Interest







Climatic Change

Diurnal

Seasonality

Interannual

Decadal

Long-term

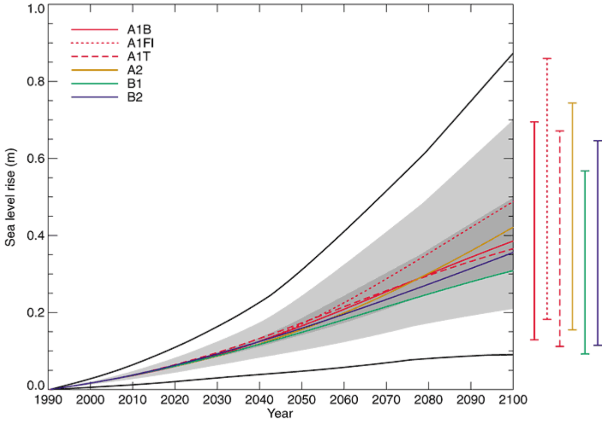
Introduction

UNIVERSIDAD DE CANTABRIA

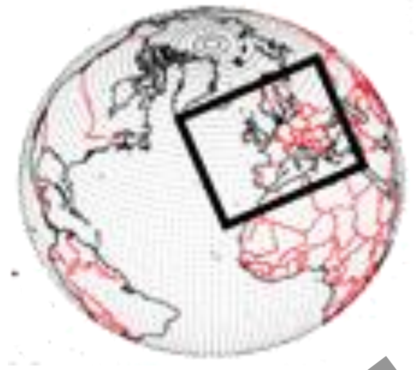
GeOcean
Geomatics and Ocean Engineering Group

Statistical downscaling approach based on weather types

Climate change scenarios

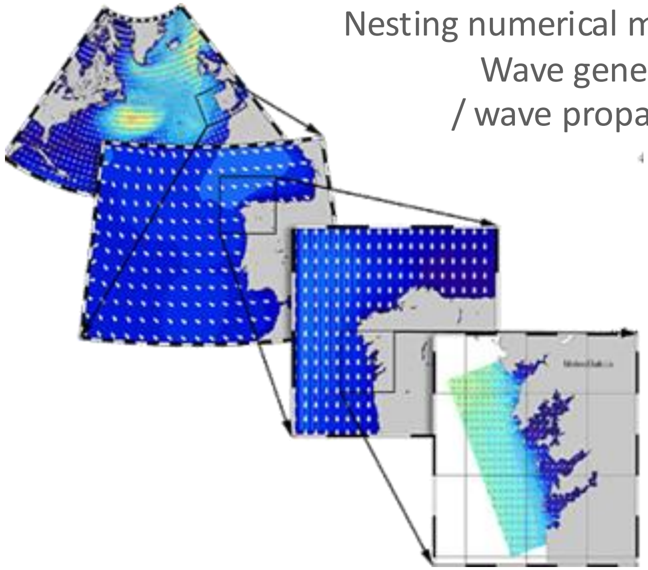


Global Circulation Model (GCM)

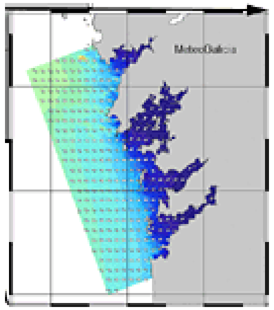


Dynamical downscaling

Nesting numerical models
Wave generation
/ wave propagation



Statistical Downscaling



Statistical methods
Relates local weather with global atmospheric predictors

Y = f (X)

Introduction

ATMOSPHERIC CIRCULATION
(predictor X: SLP)

MULTIVARIATE WAVE CLIMATE

(predictand Y: H, T, Dir)

Regional atmospheric climatology (X)

Local wave climatology (Y)

Introduction

Multivariate Extreme Events

$$\eta_{TWL} = \eta_{AT} + \eta_{SS} + 0.3 * H_s$$

Santander – Sardinero

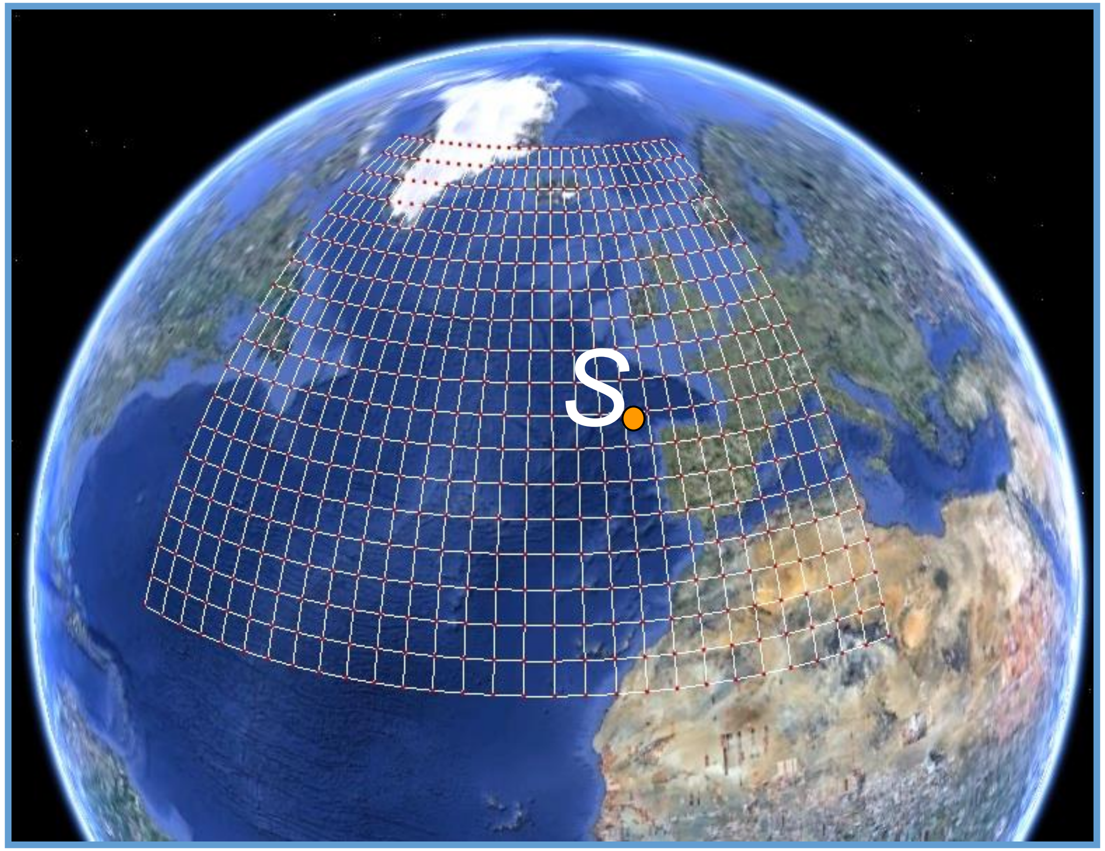
Santander – Somo
Storms 2013/14

Severe storms + spring tides drived extraordinary coastal erosion and flooding

Introduction

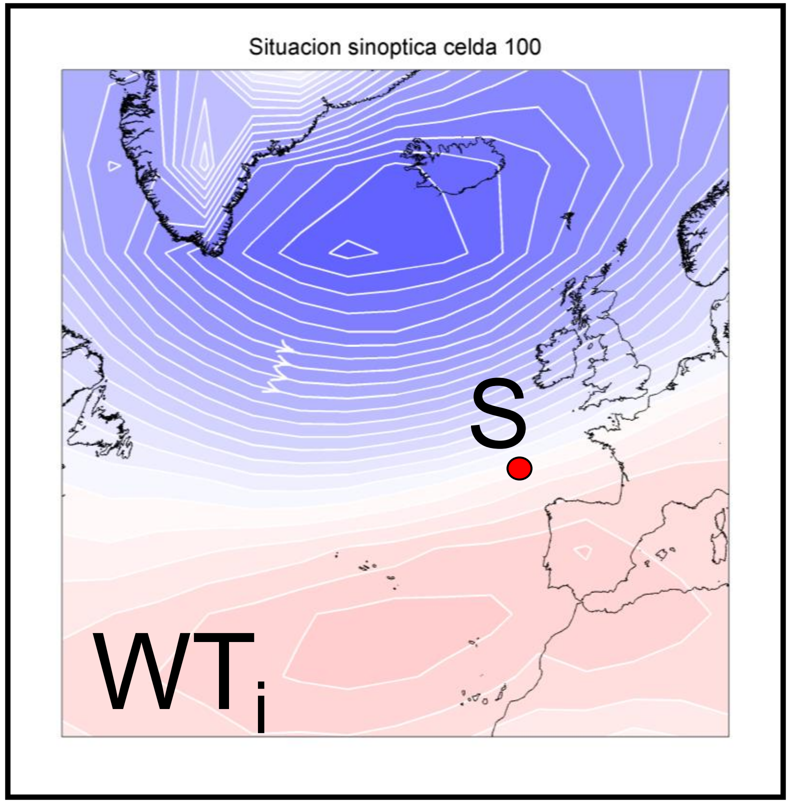
Statistical downscaling approach based on weather types

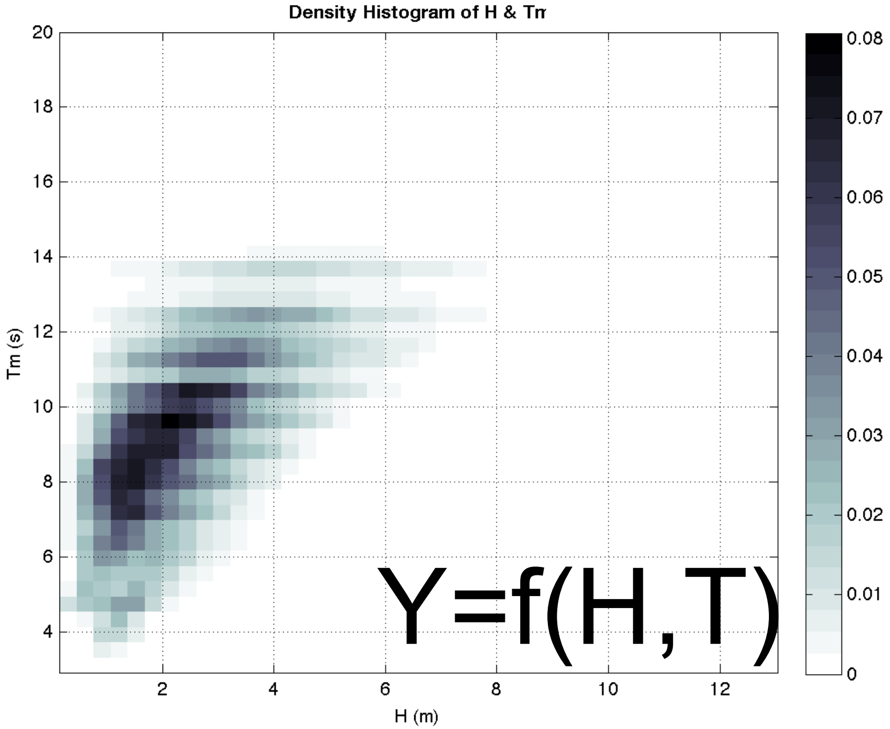
Predict multivariate wave climate (Y) at a particular location *S* as a function of synoptic atmospheric circulation (X)



Introduction

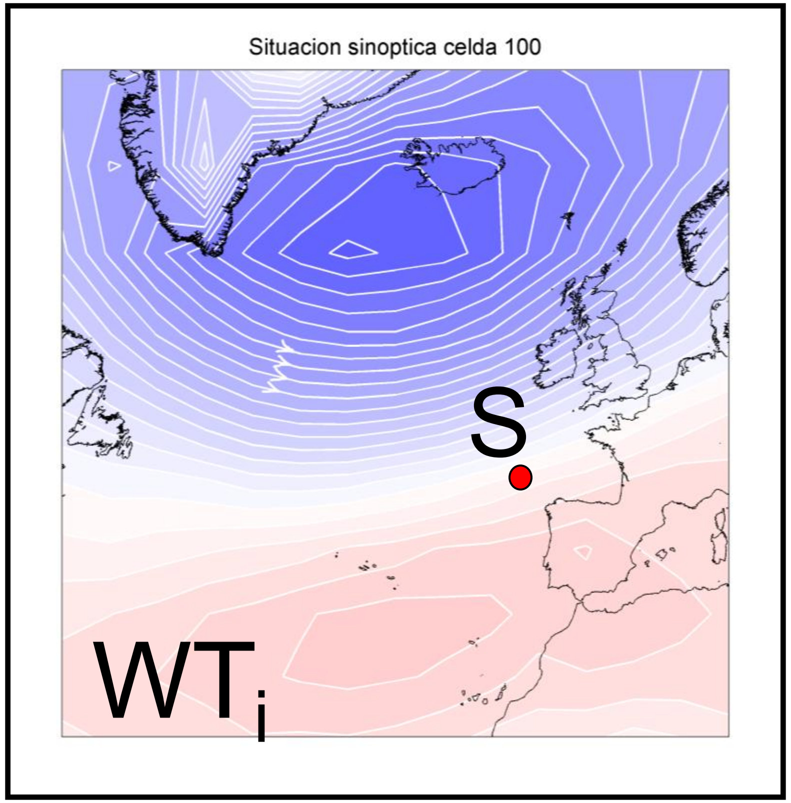
Predict multivariate wave climate (Y) at a particular location *S* as a function of synoptic atmospheric circulation patterns (X)

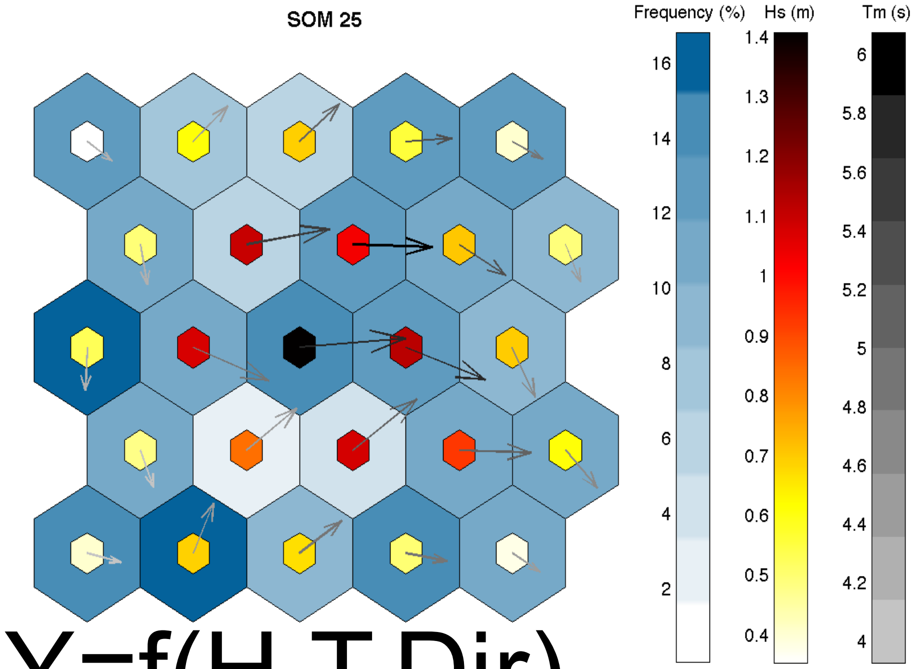




Introduction

Predict multivariate wave climate (Y) at a particular location *S* as a function of synoptic atmospheric circulation patterns (X)



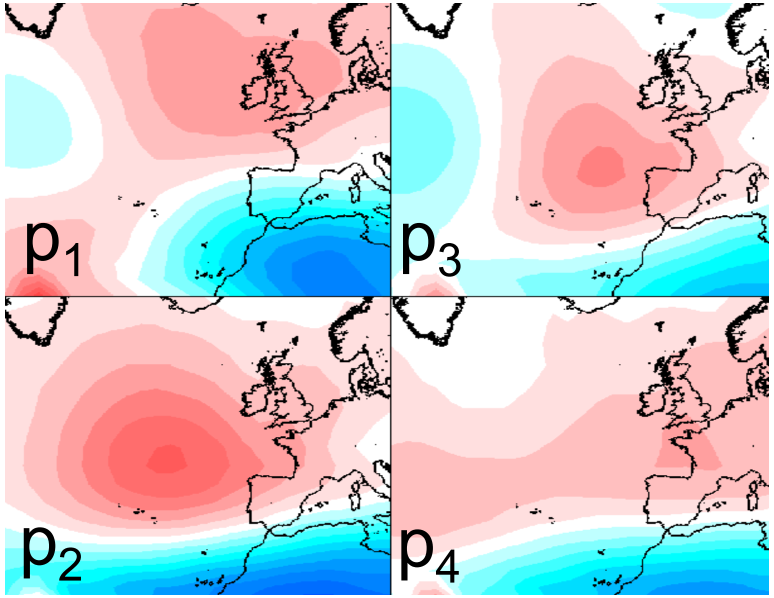


Y=f(H,T,Dir)

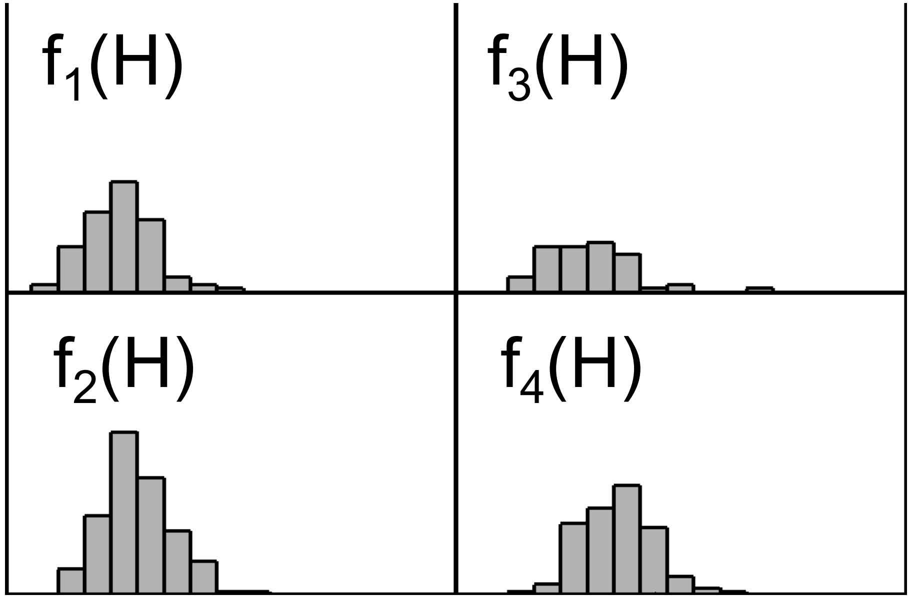
Camus et al 2011

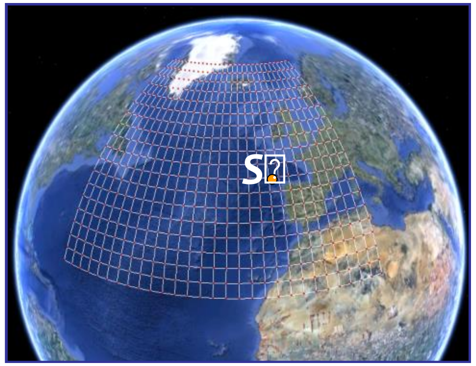
Introduction

$X=(WT_1,WT_2,WT_3,WT_4)$



Y





WT= Weather-type

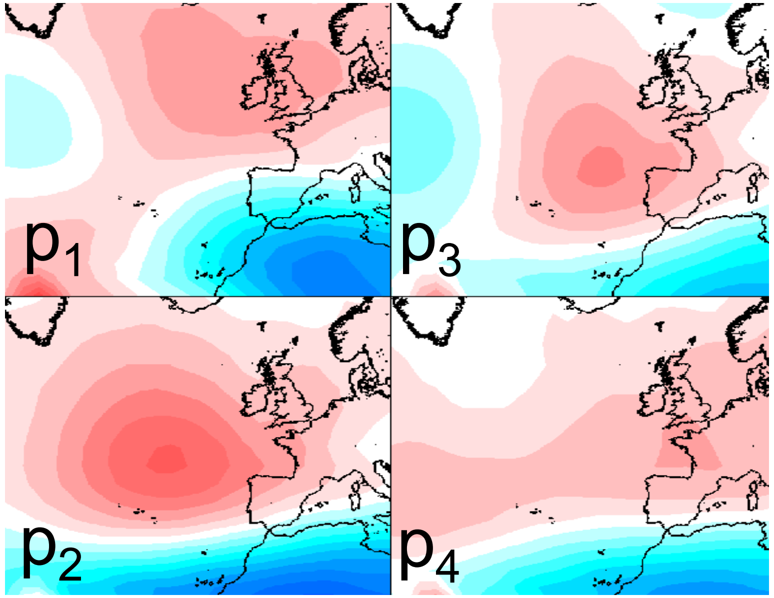
p_i=ocurrence probability of WT_i

Regression model /Stat. Downscaling:

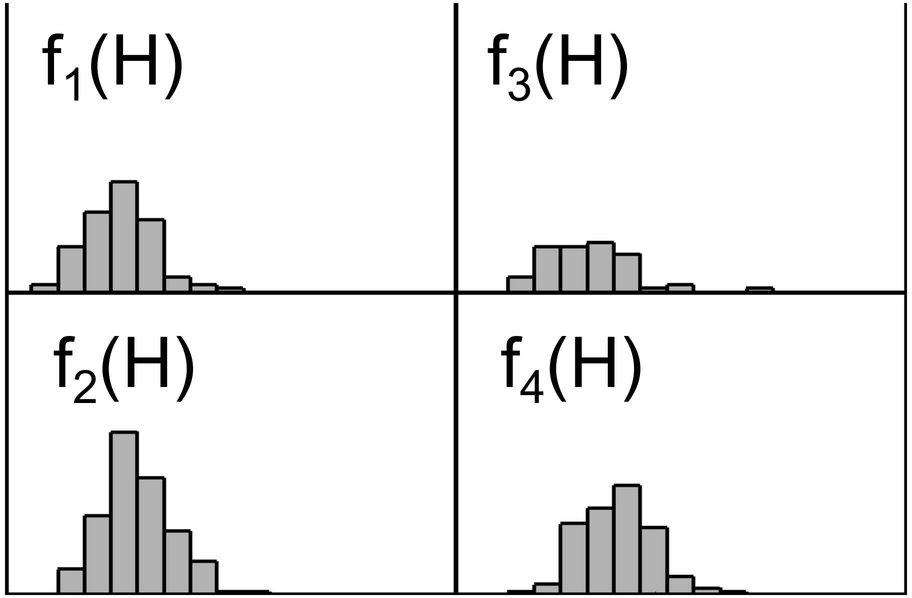
$$Y=g(X)$$

Introduction

$X=(WT_1,WT_2,WT_3,WT_4)$



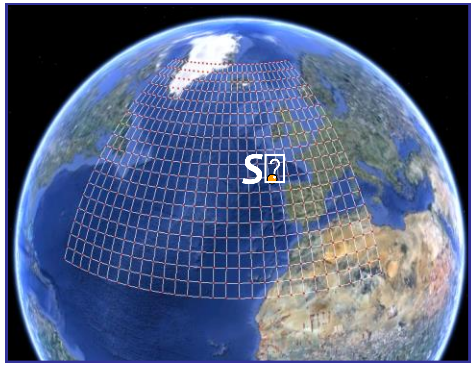
Y



p_i=ocurrence probability of WT_i

$p_1+p_2+p_3+p_4=1$

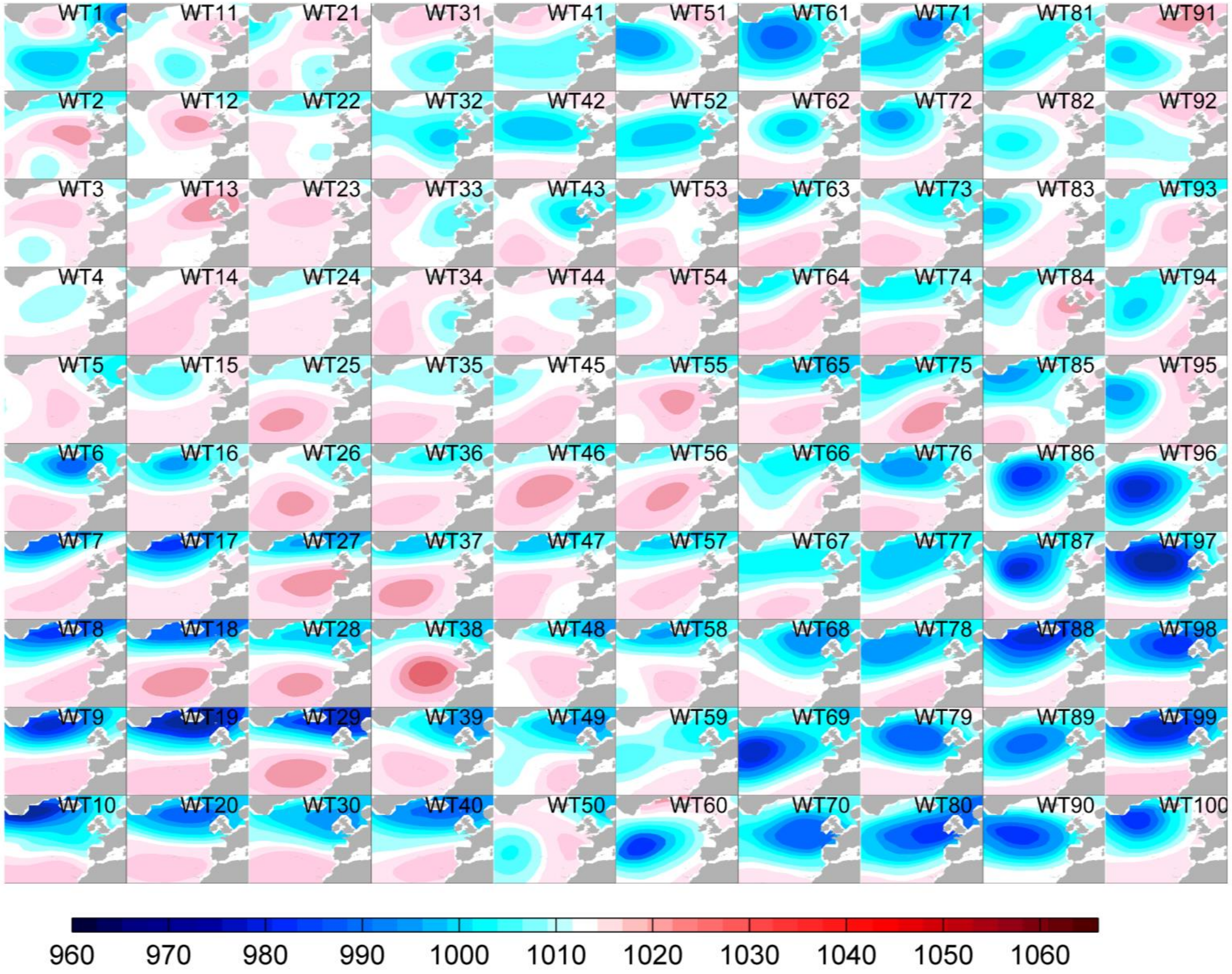
$f_S(H)=p_1f_1(H)+p_2f_2(H)+p_3f_3(H)+p_4f_4(H)$



Introduction

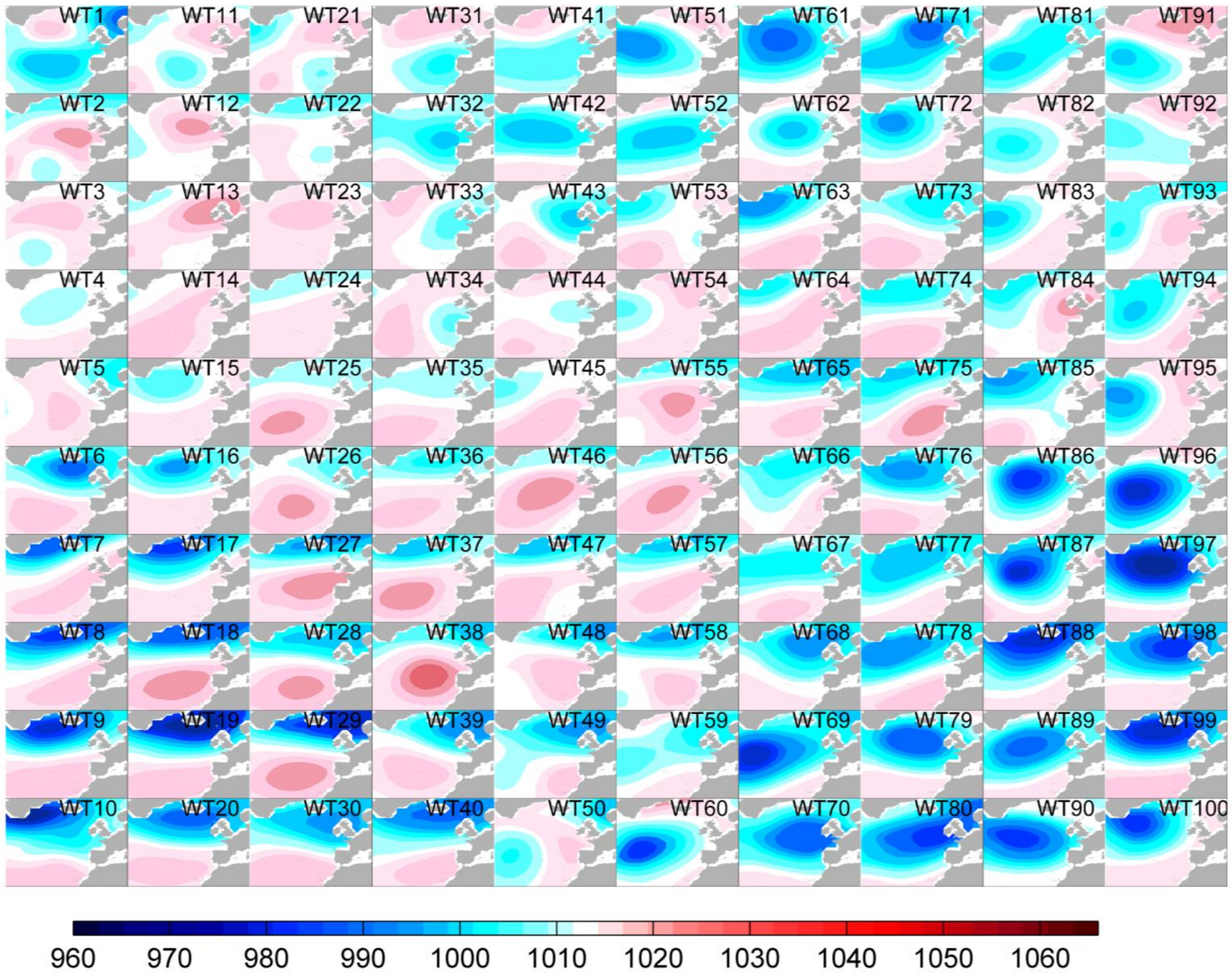
UC UNIVERSIDAD DE CANTABRIA

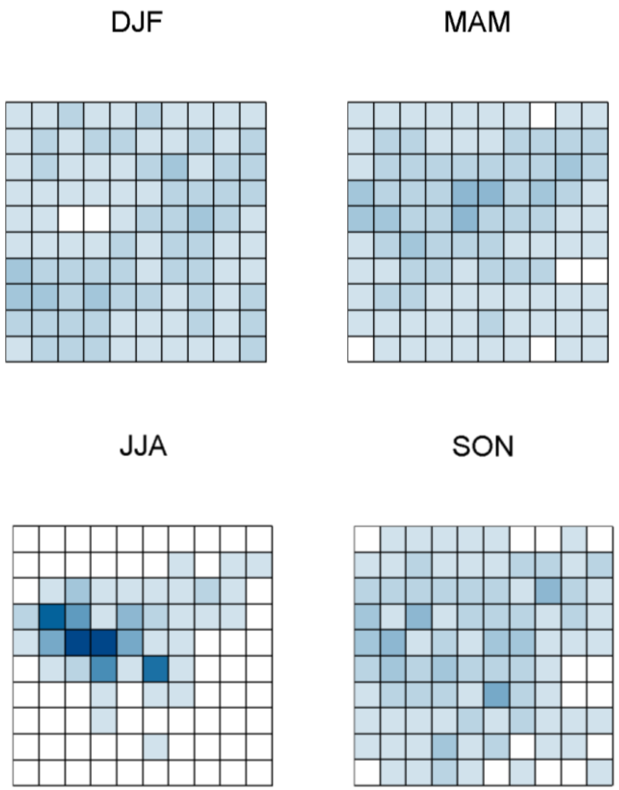
GeoOcean Geomatics and Ocean Engineering Group



Camus et al 2014

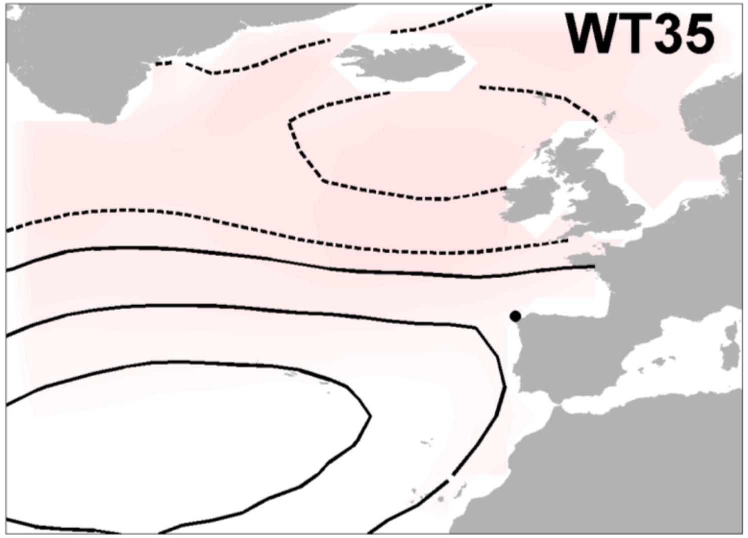
Introduction

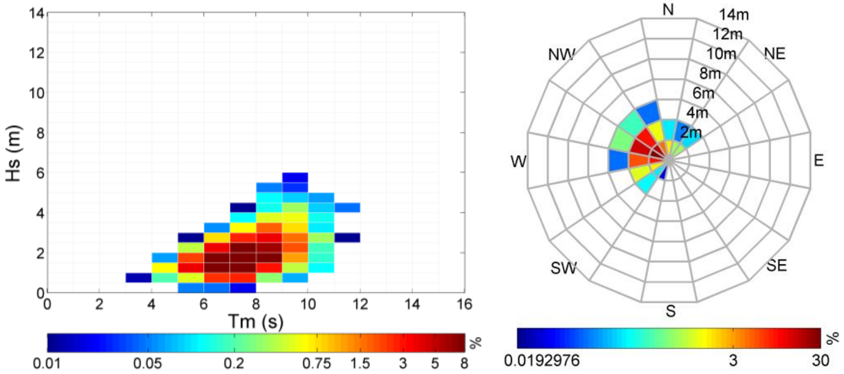




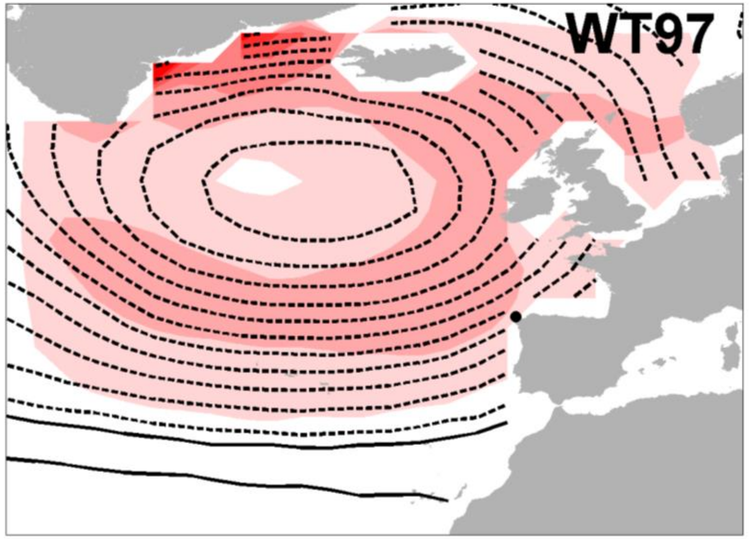
Introduction

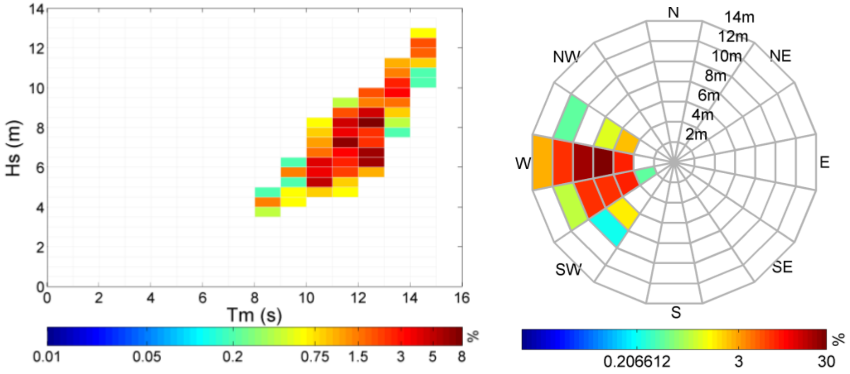
Summer pattern



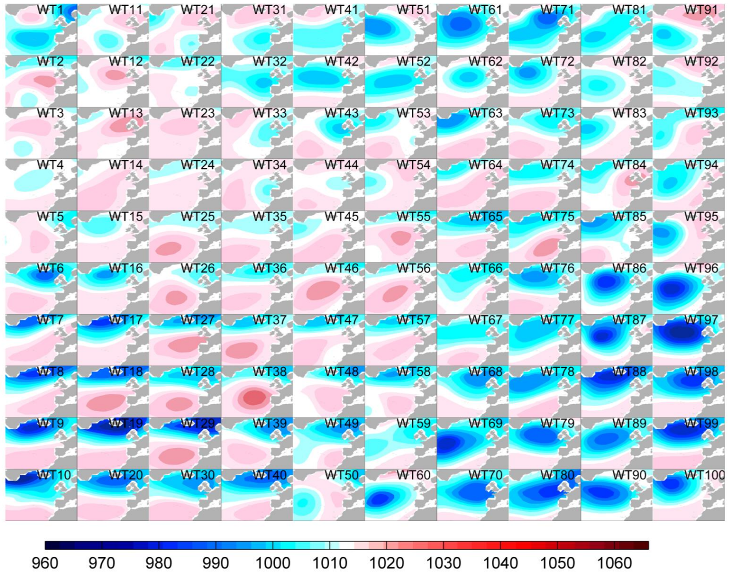


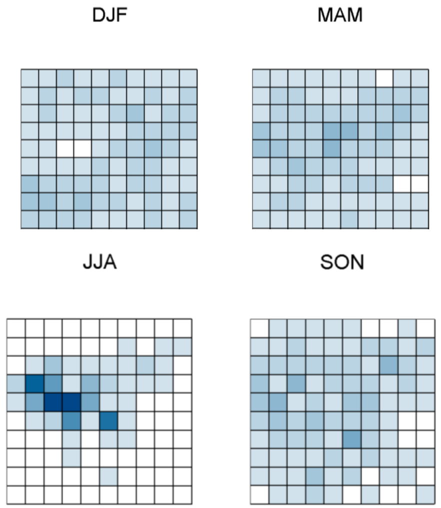
Winter pattern

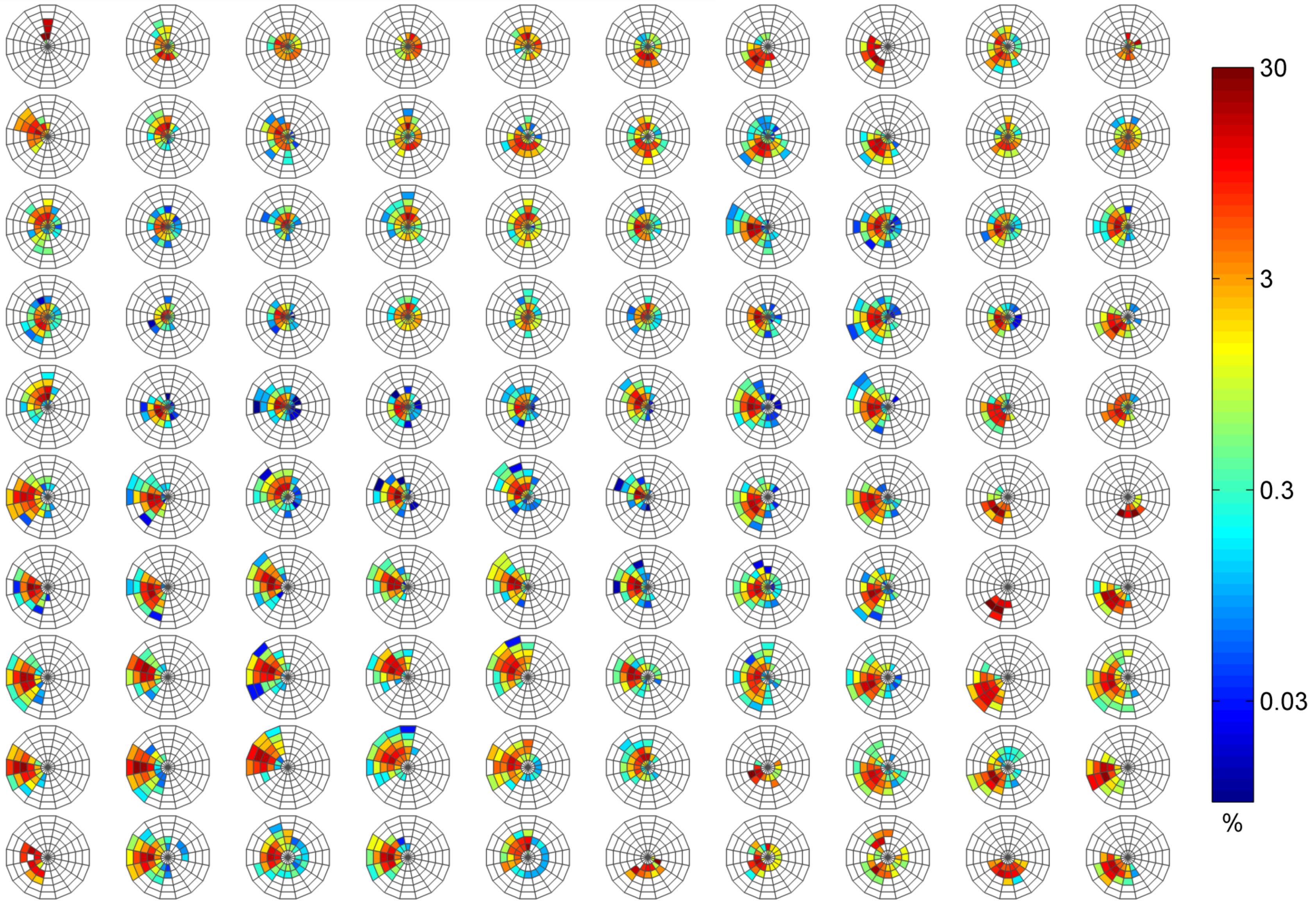




Introduction





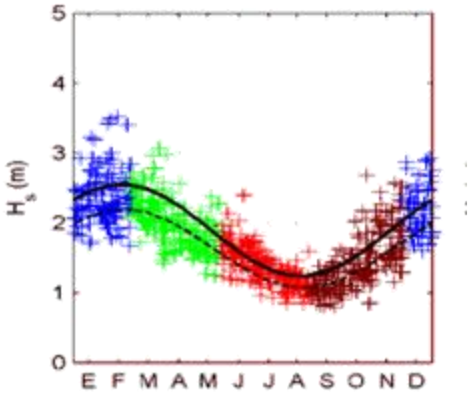


Introduction

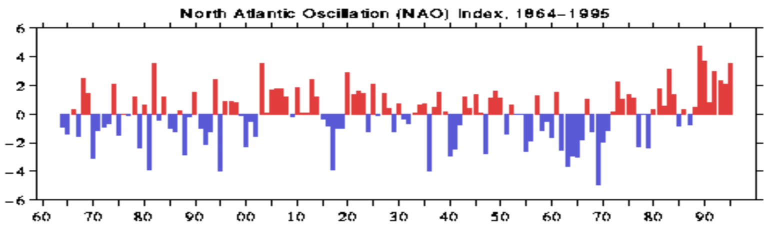
Climate non stationarity

Many weather and climate extremes are the result of **natural climate variability** (including phenomena such as El Niño), and natural decadal or multi-decadal variations in the climate provide the backdrop for **anthropogenic climate changes**.

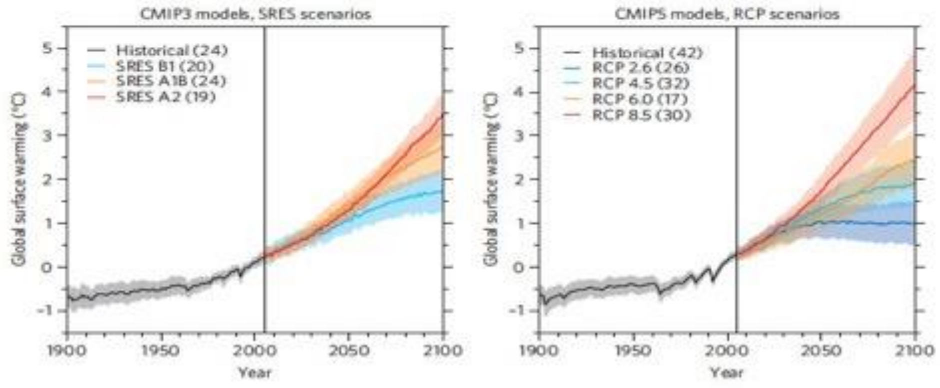
A **changing climate leads** to changes in the frequency, intensity, spatial extent, duration, and timing of weather and climate extremes, and can result in **unprecedented extremes**.



Seasonality



Interannual variability

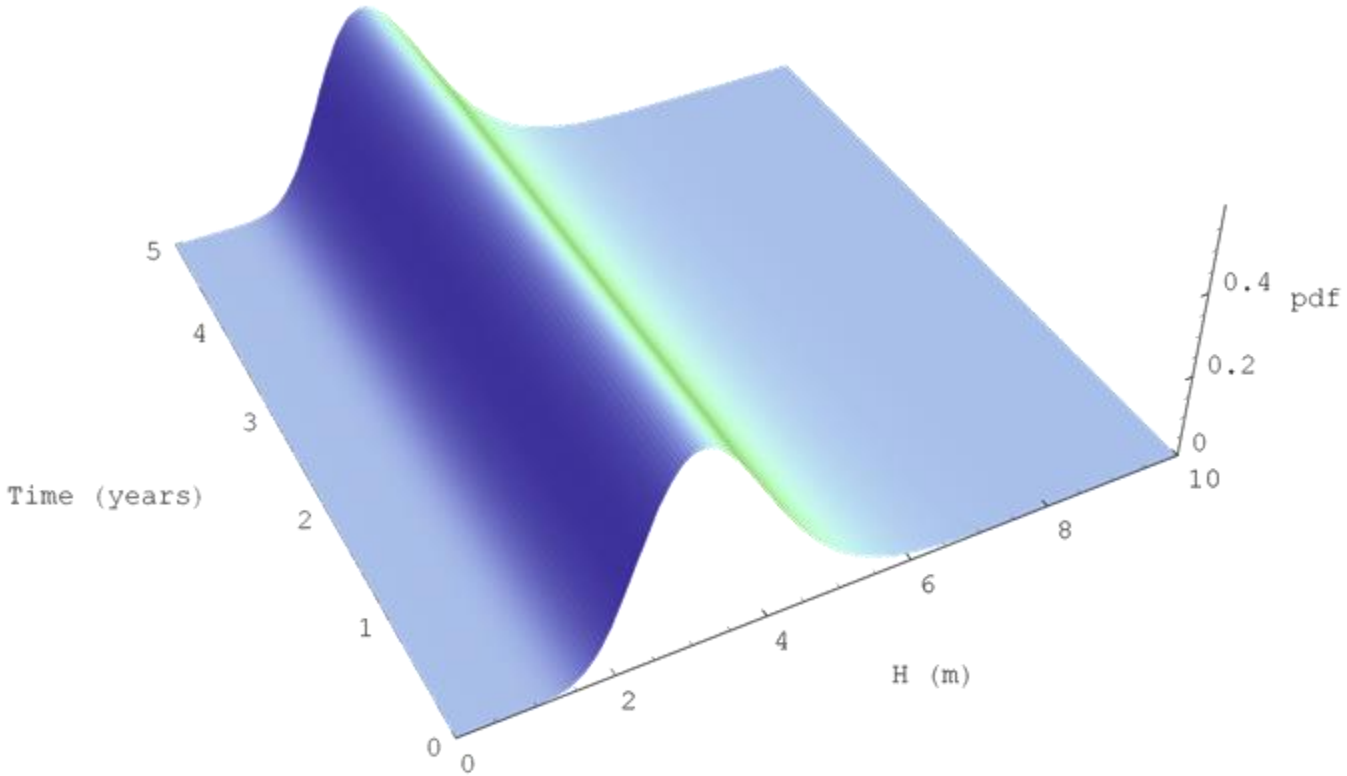


Climate change

Introduction

Statistical methods of extremes

STATIONARY CASE



$$\mu = \mu_0$$

$$\psi = \psi_0$$

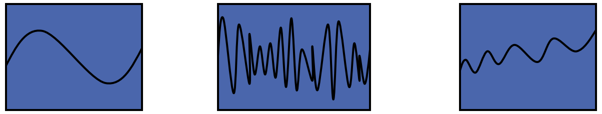
$$\xi = \xi_0$$

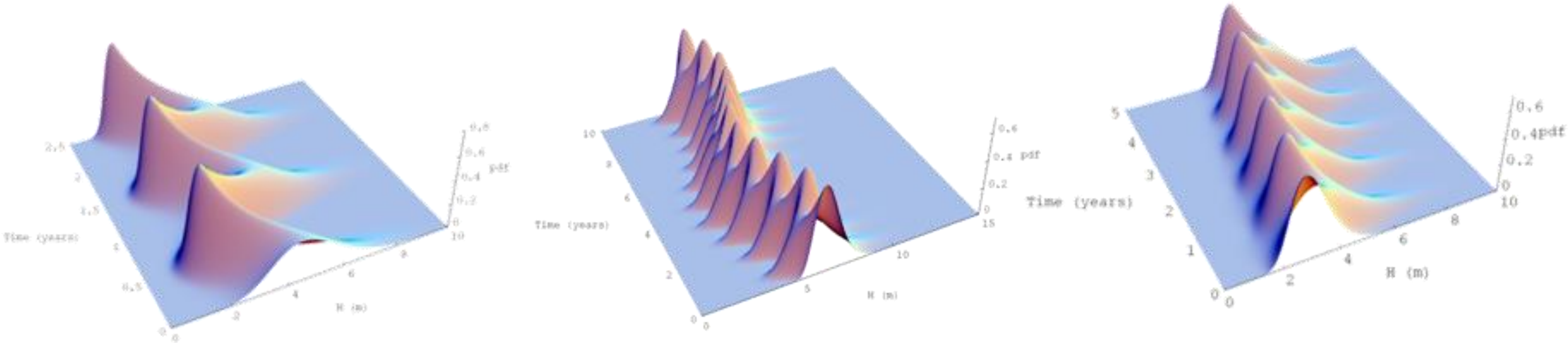
Izaguirre, 2011

Introduction

Statistical methods of extremes

NON-STATIONARY CASE





$$\mu = \mu(t)$$

Parametric models $\longrightarrow$ $$\psi = \psi(t)$$

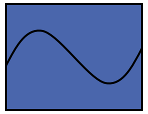
$$\xi = \xi(t)$$

Izaguirre, 2011

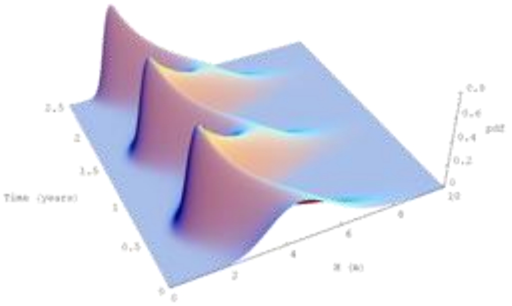
Introduction

Statistical methods of extremes

NON-STATIONARY CASE



SEASONALITY



Méndez et al. (2006)

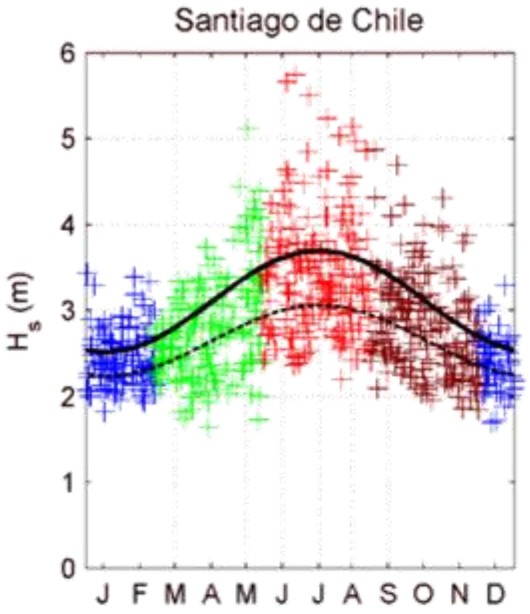
Menéndez et al. (2008)

Menéndez et al. (2009)

Hemer (2010)

$$\theta_i(t) = \beta_0 + \sum_{i=1}^{P} \left[\beta_{2i-1} \cos(ikt) + \beta_{2i} \sin(ikt) \right]$$

$$\begin{cases} \mu(t) = \beta_0 + \beta_1 \cos(2\pi t) + \beta_2 \sin(2\pi t) \\ \psi(t) = \alpha_0 + \alpha_1 \cos(2\pi t) + \alpha_2 \sin(2\pi t) \\ \xi(t) = \gamma_0 \end{cases}$$

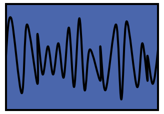


Izaguirre, 2011

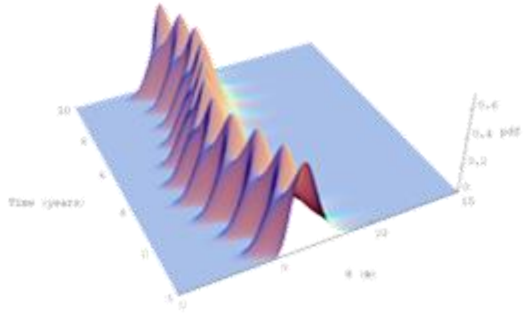
Introduction

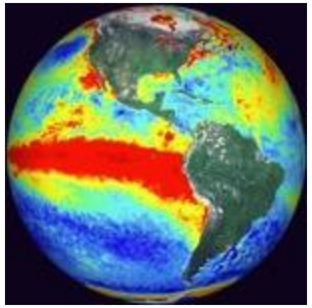
<u>Statistical methods of extremes</u>

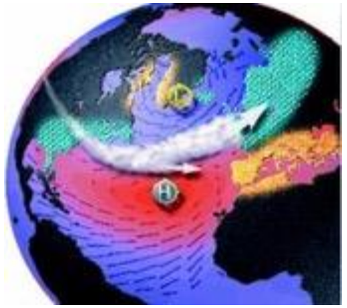
NON-STATIONARY CASE



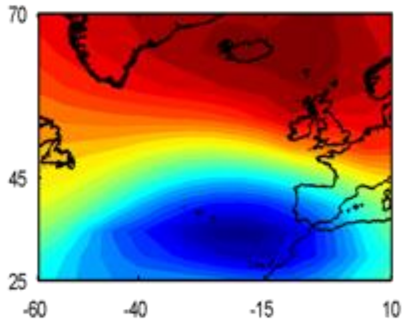
INTERANNUAL VARIABILITY







...



Woolf y Challenor (2002)

Lionello y Sanna (2005)

Méndez et al. (2006)

Menéndez et al. (2008)

Hemer et al. (2010)

$$\boxed{\theta_i(t) = \beta_0 + \beta_i \cdot covariable(t)}$$

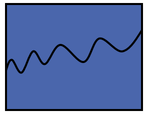
$$\begin{cases} \mu(t) = \beta_0 + \beta_1 \cos(2\pi t) + \beta_2 \sin(2\pi t) + \beta_{Niño3} Niño3(t) \\ \psi(t) = \alpha_0 + \alpha_1 \cos(2\pi t) + \alpha_2 \sin(2\pi t) \\ \xi(t) = \gamma_0 \end{cases}$$

Izaguirre, 2011

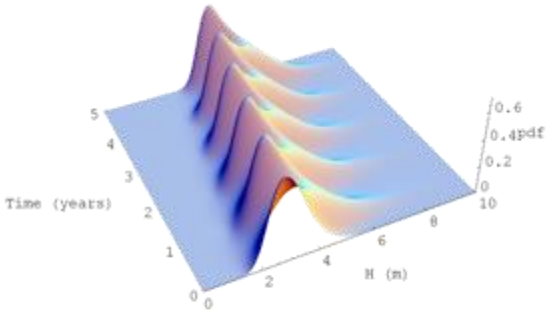
Introduction

<u>Statistical methods of extremes</u>

NON-STATIONARY CASE



LONG-TERM TREND



Gulev y Grigorieva (2006)

Menéndez et al. (2008)

Hemer (2010)

$$\boxed{\theta_i(t) = \beta_0 \cdot e^{\beta_1 t} \approx \beta_0 \cdot (1 + \beta_1 t)}$$

$$\begin{cases} \mu(t) = \beta_0 + \beta_1 \cos(2\pi t) + \beta_2 \sin(2\pi t) + \beta_{LT} \cdot t \\ \psi(t) = \alpha_0 + \alpha_1 \cos(2\pi t) + \alpha_2 \sin(2\pi t) \\ \xi(t) = \gamma_0 \end{cases}$$

Izaguirre, 2011

Introduction

UNIVERSIDAD DE CANTABRIA

GeoOcean Geomatics and Ocean Engineering Group

State of knowledge

Analysis of boundary conditions – Extreme analysis

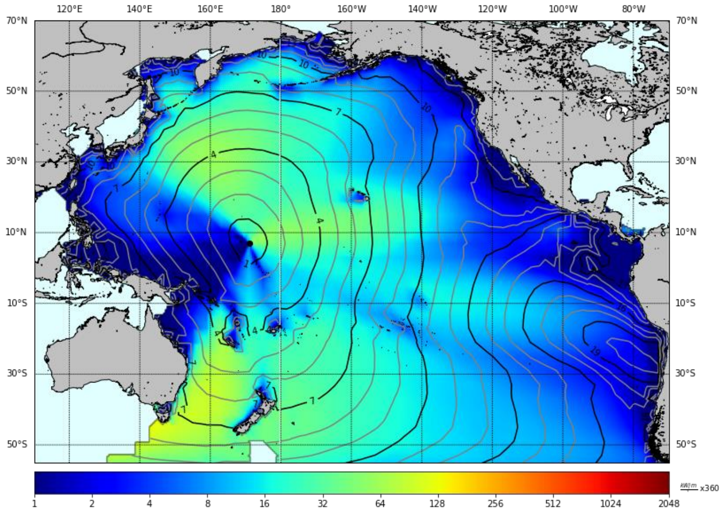
	Stationary		**Non – Stationary**	
Univariate	Traditional approach	Coles, 2001 GEV POT	Katz et al. 2002 Méndez et al. 2006 Cannon 2010 …	Climate as a covariate
Multivariate	Hawkes et al. 2002 Defra 2005 Gouldby et al. 2008	Heffernan & Tawn 2004 Wahl et al. 2012 …	Bender et al. 2014 Masina et al. 2015	

Simplifying assumptions on the dependence structure

Bivariate copulas

Framework

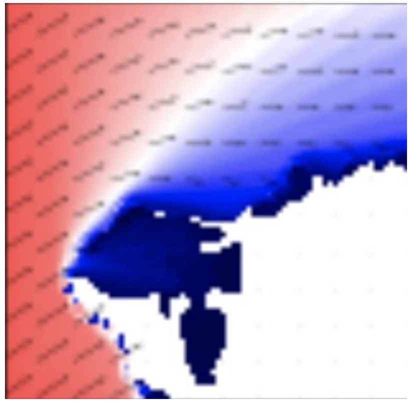
Large-scale met-ocean predictor



X

Sea Level Pressure (SLP)
10-m wind (U10)
Sea Surface Temperature (SST)

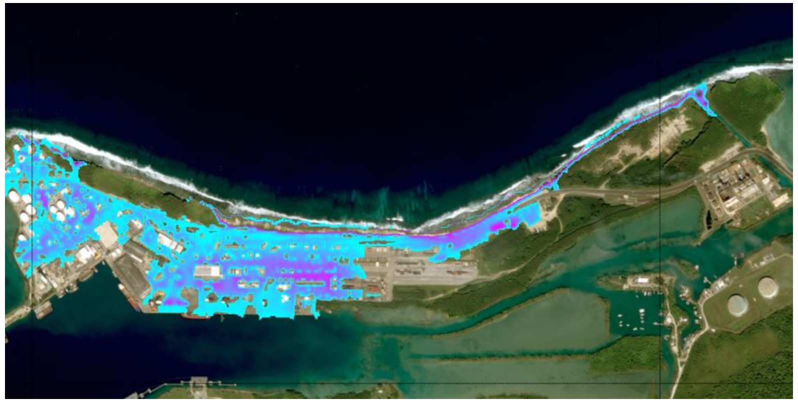
Local hydro-met-ocean dynamics



Y

Waves
Storm surge
Wind
Sediment discharge

Local coastal impacts



Z

Flooding
Erosion/Accretion
Reliability
Risk assessment

Framework

Univariate	$X = X_1 = X$	Univariate pdf	$f_x(x)$
Multivariate	$X = (X_1, X_2, \ldots, X_n)$	Multivariate pdf	$f_x(x) = f_x(x_1, x_2, \ldots x_{Nx})$

Large-scale met-ocean predictor **Local hydro-met-ocean dynamics**

$$X \longrightarrow Y$$

Dynamical Downscaling	$X \xrightarrow{DD} Y$
Statistical Downscaling	$X \xrightarrow{SD} Y$
Hybrid Downscaling	$X \xrightarrow{HD} Y$

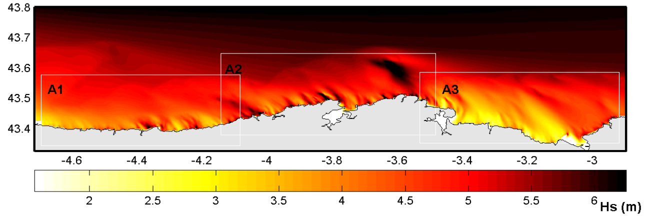
$$Y = f(X)$$

Framework

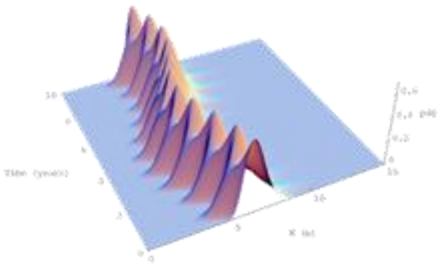
UNIVERSIDAD DE CANTABRIA

GeoOcean Geomatics and Ocean Engineering Group

$X \xrightarrow{DD} Y$

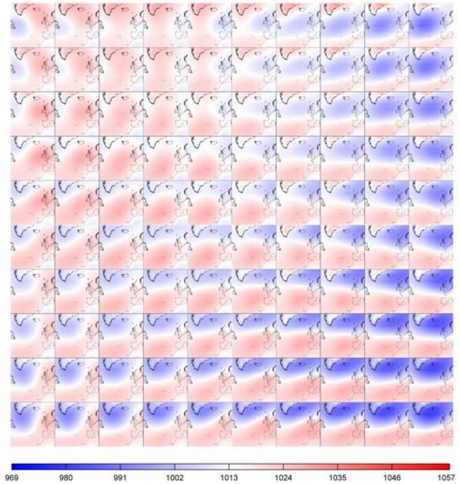


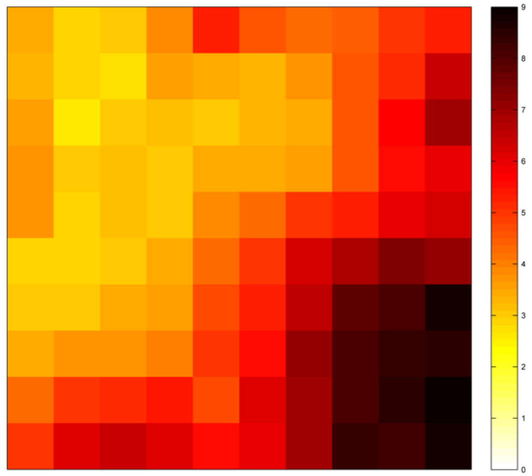
$X \xrightarrow{SD} Y$



$X \xrightarrow{HD} Y$

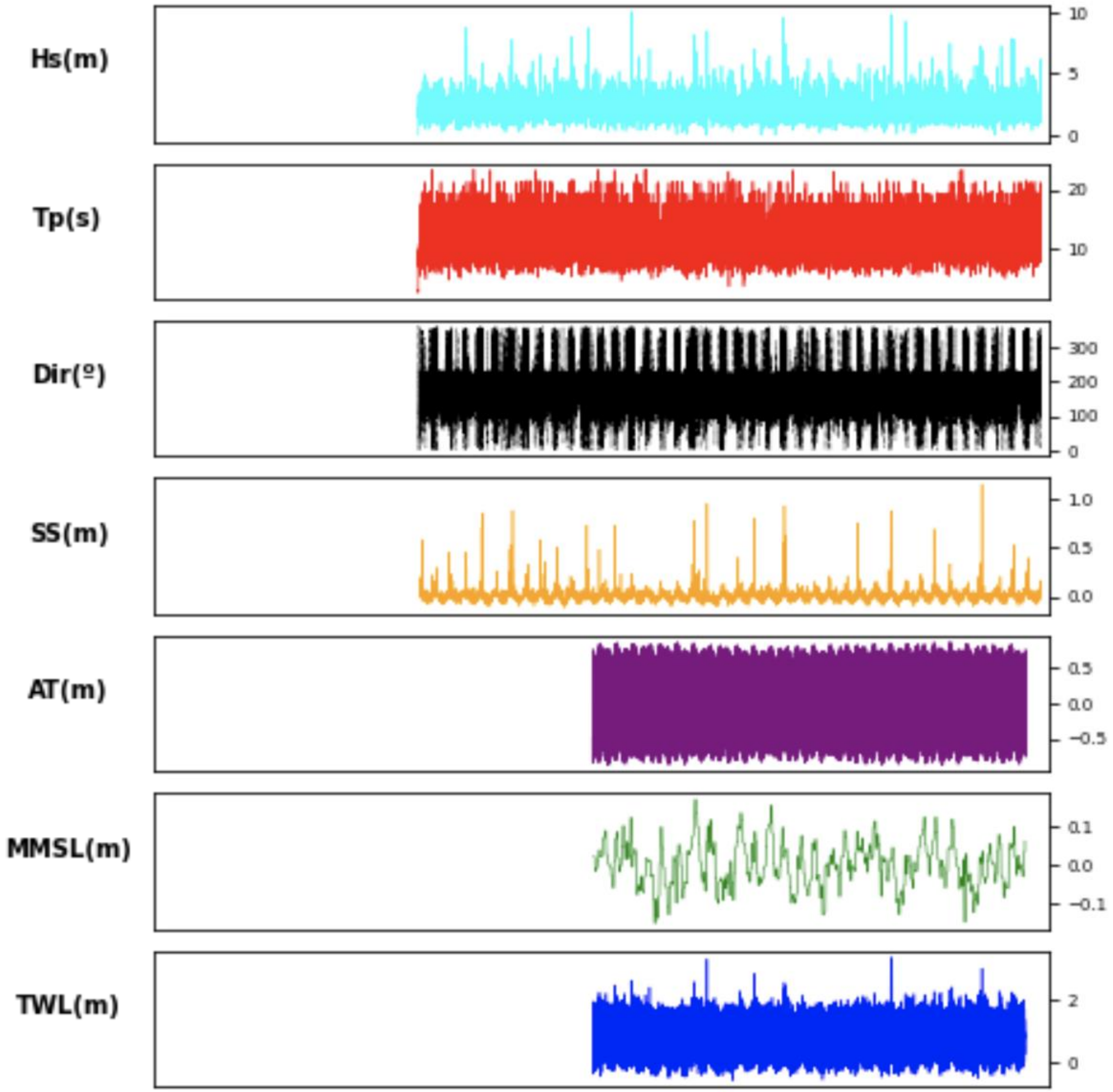
$X \xrightarrow{SD} Y$





Nomenclature

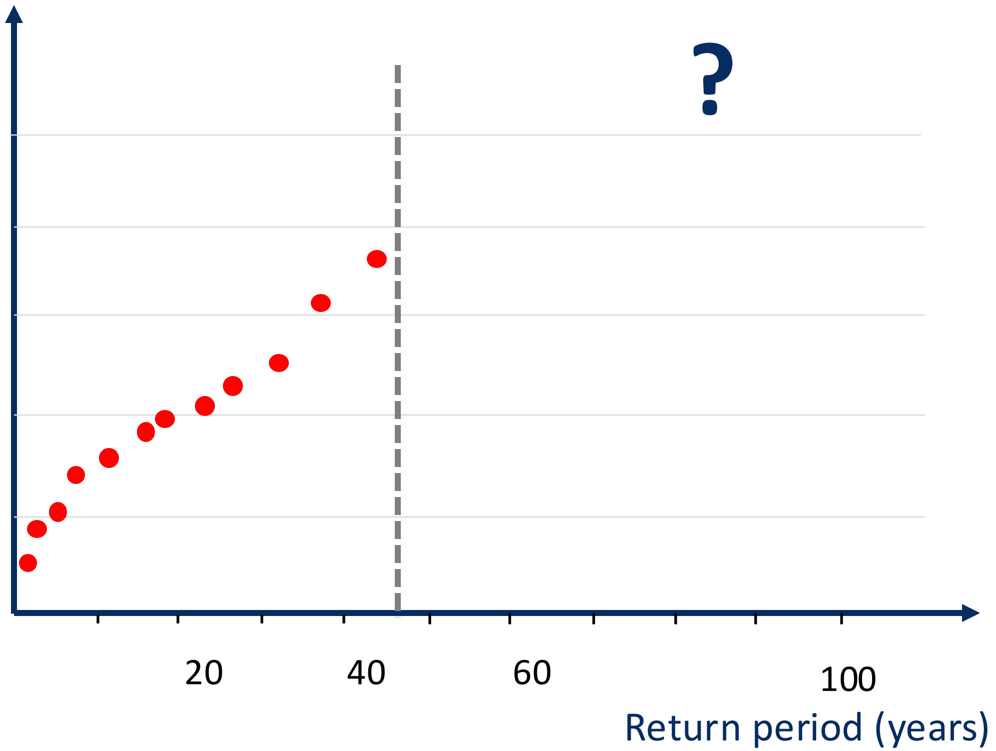
Extreme Value Analysis

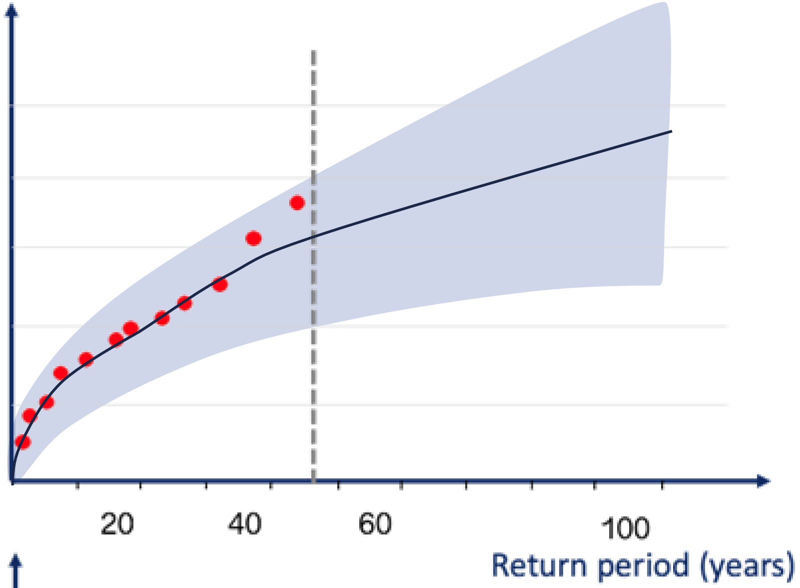


Limited number of record years

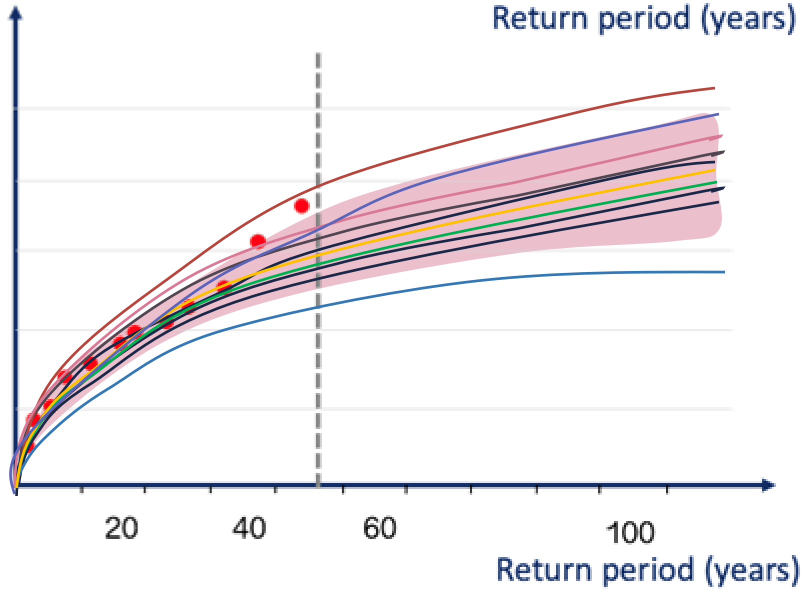
Limited number of extreme events to correctly analyze a multivariate problem

Extreme Value Analysis



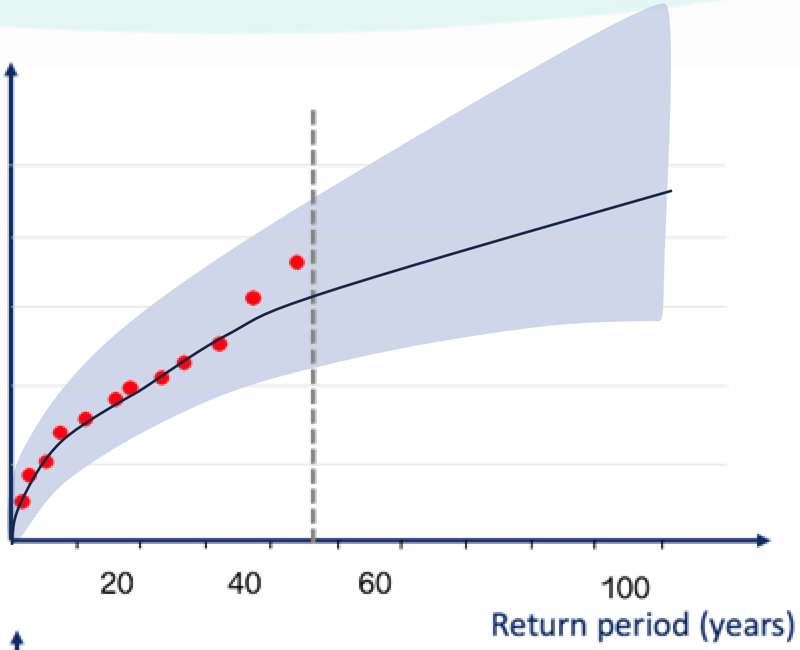


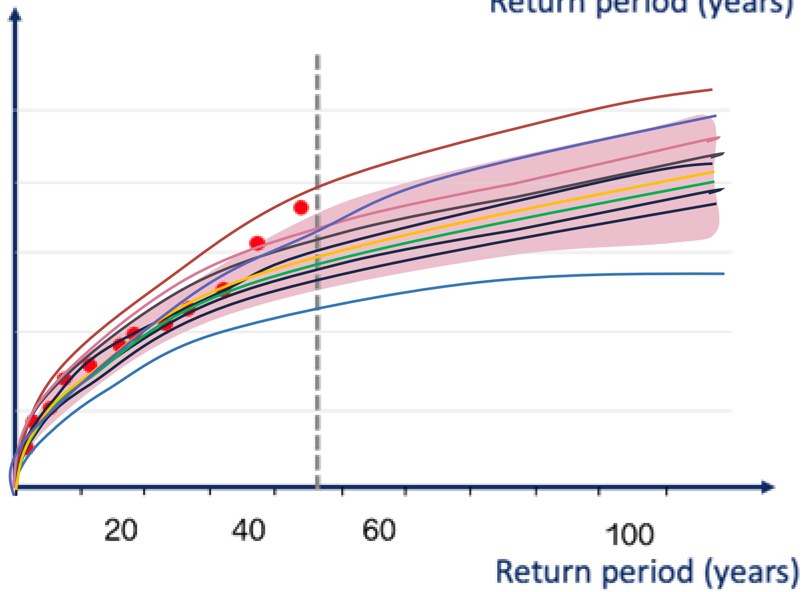
Extreme Value Fitting



Generation of synthetic time series

Extreme Value Analysis





Extreme Value Analysis

- Easy to implement
- Misguiding uncertainty: highly dependent on a reduced number of historical extreme events
- Only allows for the analysis of independent extreme values

Generation of long synthetic time series

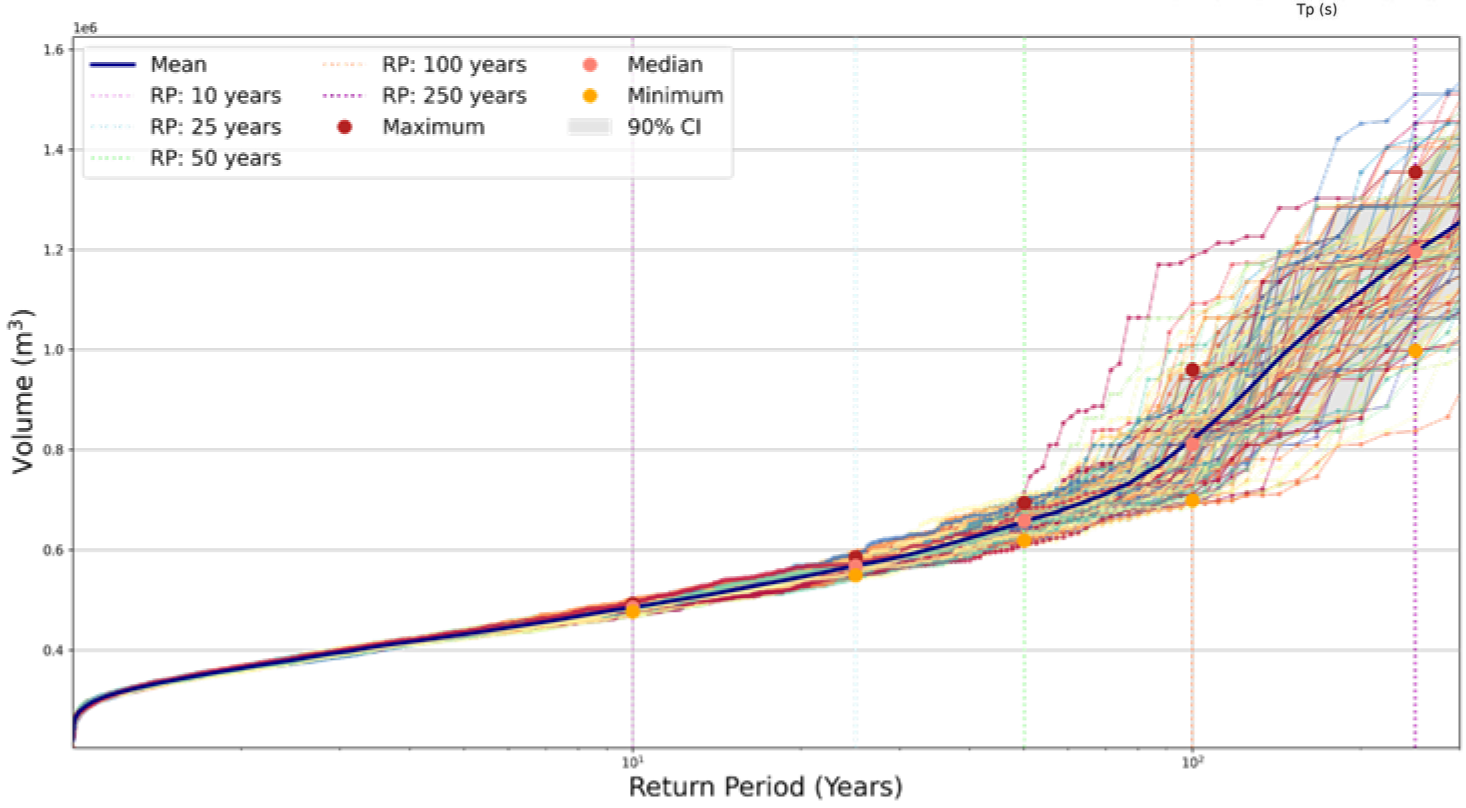
- Real exploration of the **multivariate parameter space** (Hs, Dir, AT, SS, Rain…)
- Analyze **continuous time series** (number of extreme events per year, erosion, consecutive flooding events)
- Link the events with its **climatic driver** (ENSO, Month)

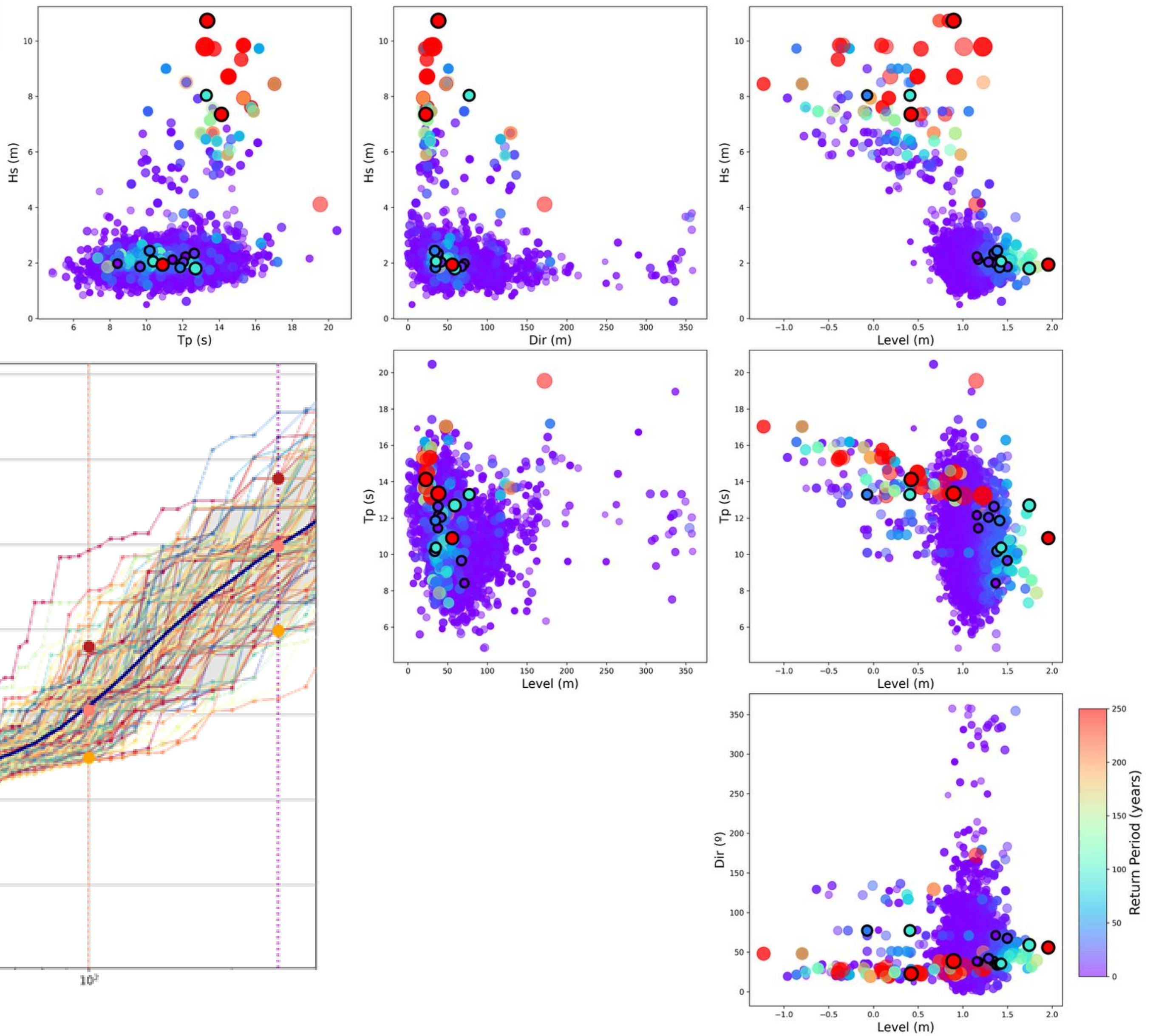
Synthetic time series

Explore Uncertainty

Example for Majuro, RMI

Unique time series with different chronologies and exploration of the multivariate space

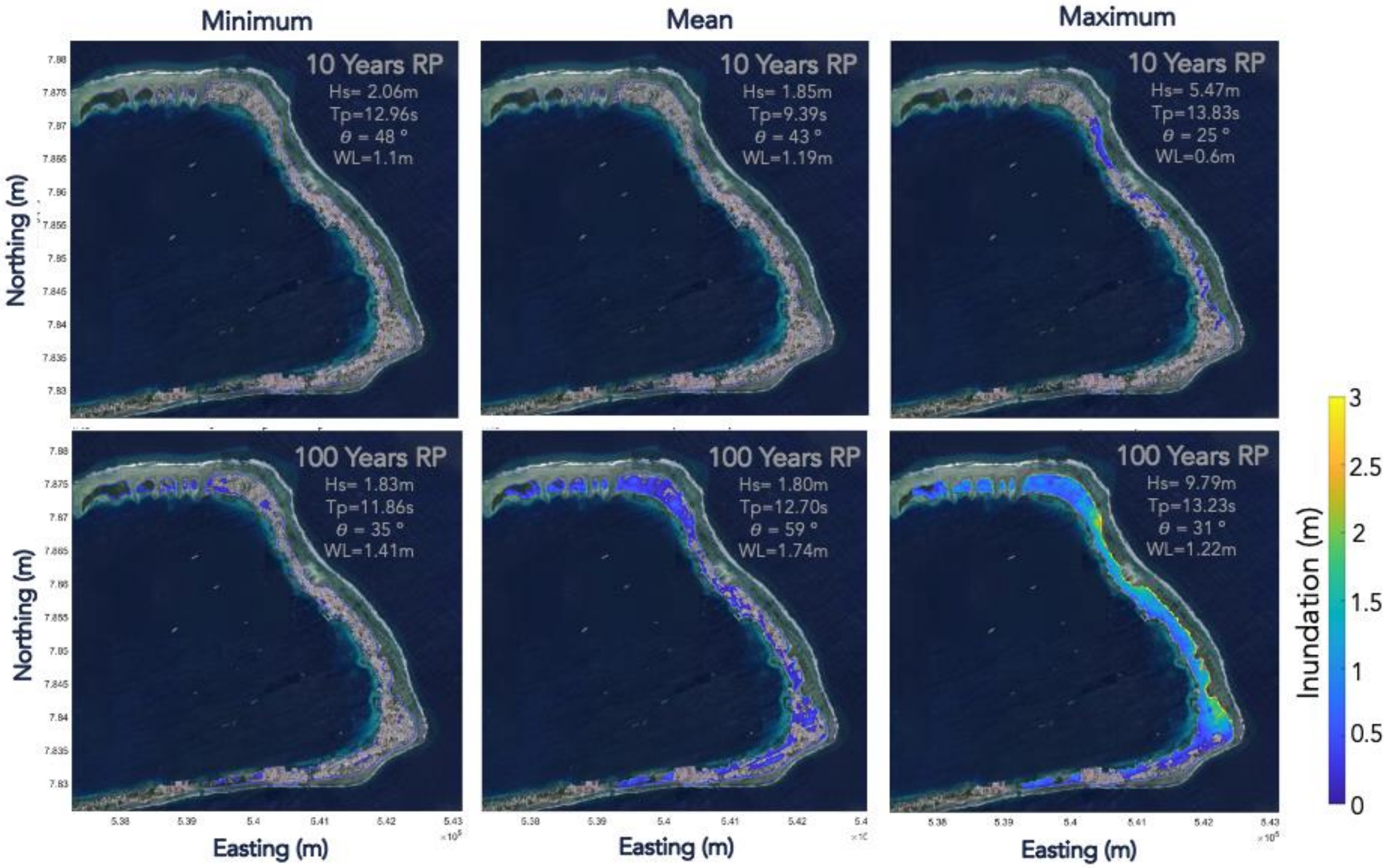




Extreme Value Analysis

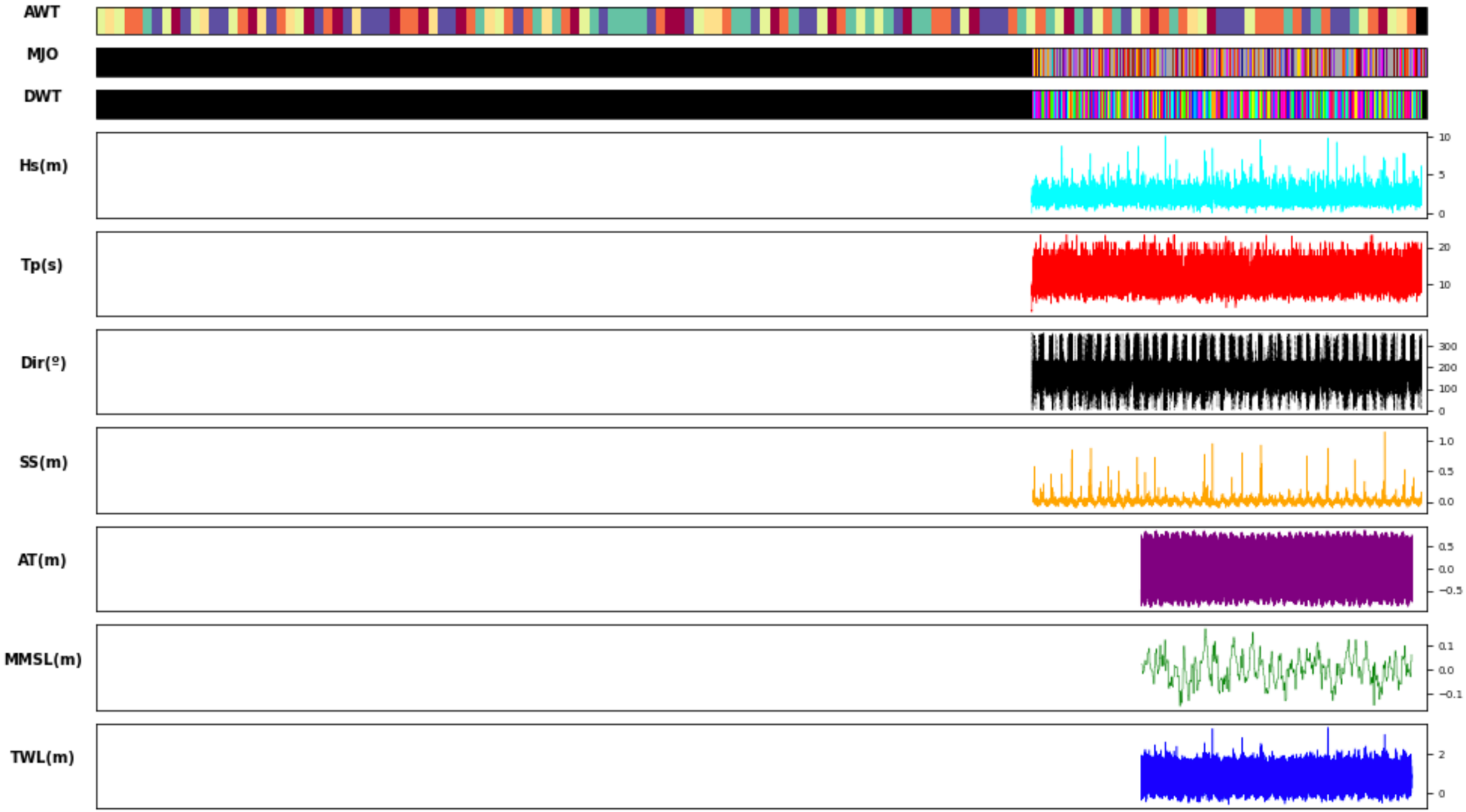
RP = 10y

RP = 100y



Climate-based Emulators

UC UNIVERSIDAD DE CANTABRIA
GeoOcean Geomatics and Ocean Engineering Group



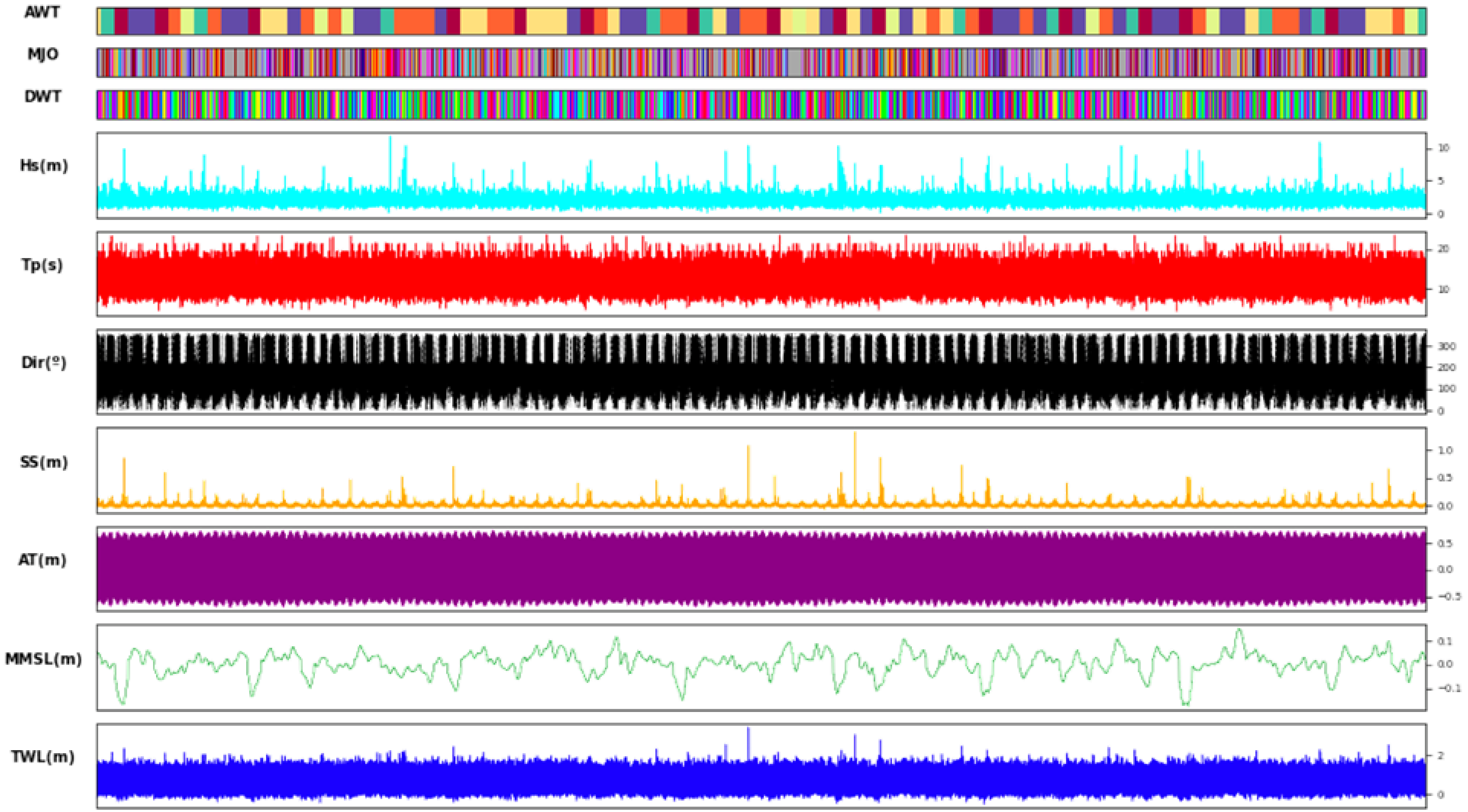
Climate-based Emulators

UNIVERSIDAD DE CANTABRIA

GeoOcean Geomatics and Ocean Engineering Group

1 Simulation x 100 Years

TESLA

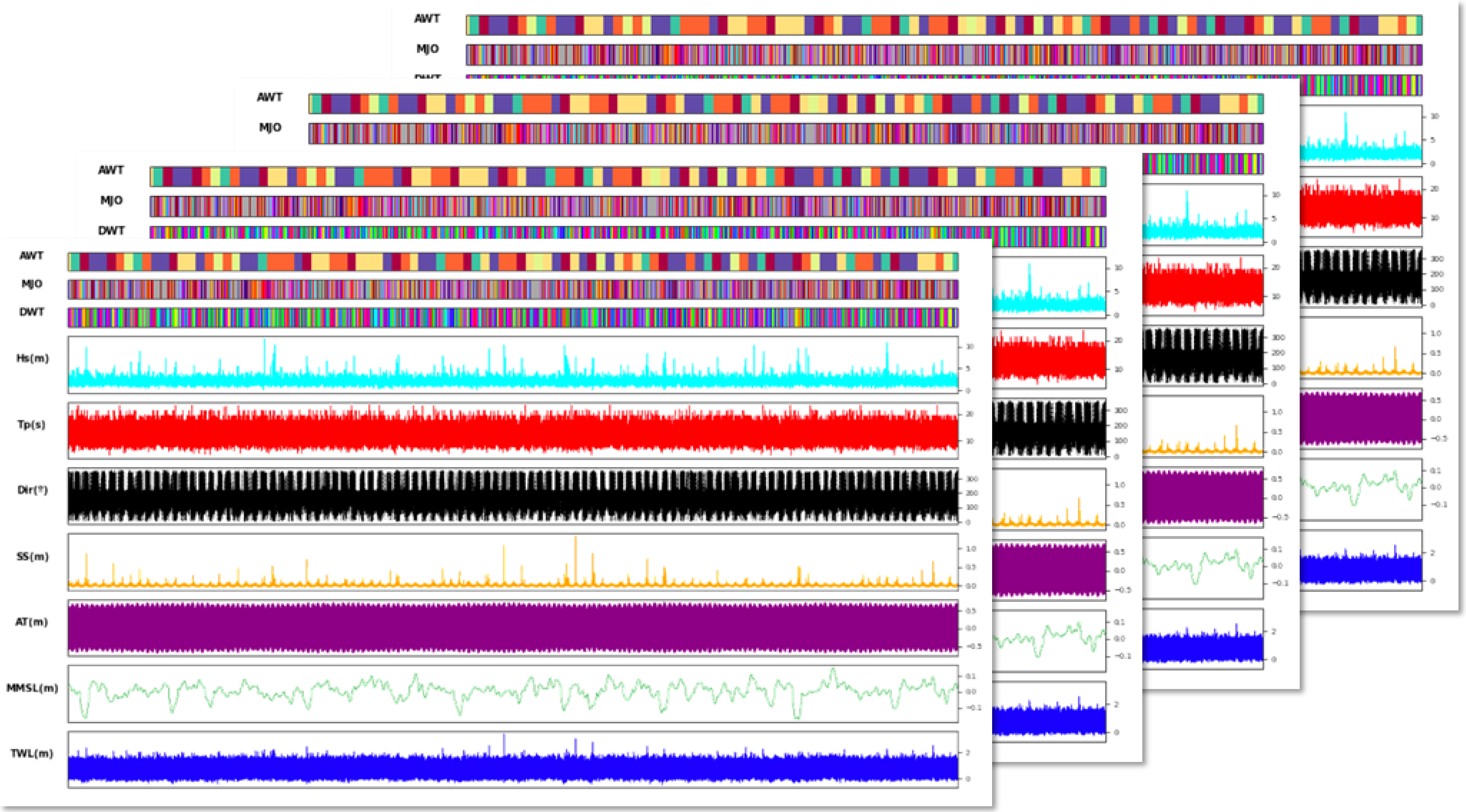


Climate-based Emulators

UNIVERSIDAD DE CANTABRIA

GeoOcean
Geomatics and Ocean Engineering Group

100s Simulations x 100s Years

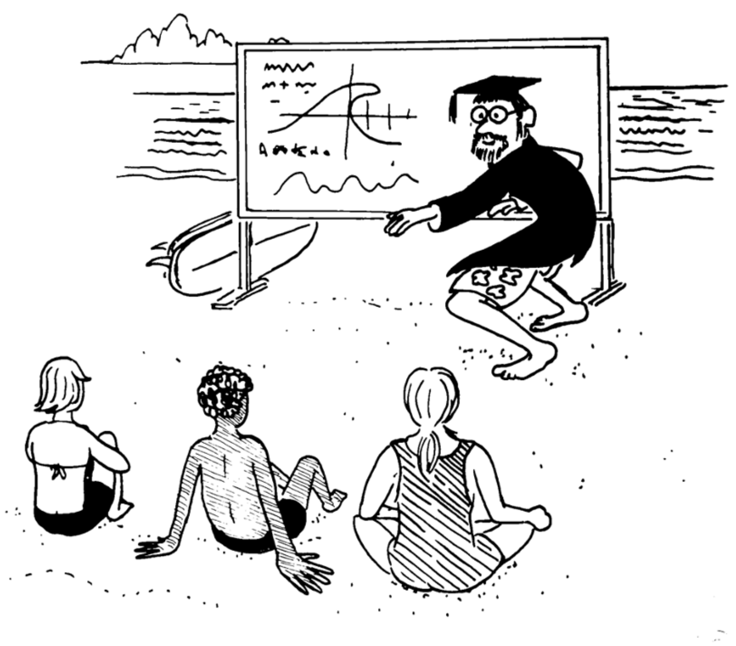


Practical classes

UC UNIVERSIDAD DE CANTABRIA

GeoOcean Geomatics and Ocean Engineering Group

BlueMath



@Image: SURFING ILLUSTRATED - A visual guide to Wave Riding. Robison, John

https://github.com/GeoOcean/BlueMath/tree/course/procoast25

UNIVERSIDAD DE CANTABRIA

GeoOcean
Geomatics and Ocean Engineering Group

Thank you

jared.ortizangulo@unican.es

https://geoocean.sci.unican.es/hyweb/

GeoOcean

@GeoOcean_UC